

國立陽明交通大學
電子物理學系
碩士論文

Department of Electrophysics
National Yang Ming Chiao Tung University
Master Thesis

二維材料於純量與向量渦旋光激發下之激子躍遷理論研究
Theoretical studies of excitonic transitions in two dimensional
materials under photoexcitation of scalar and vector vortex
beams

研究生：劉人瑩（Liu, Ren-Ying）

指導教授：鄭舜仁（Cheng, Shun-Jen）

中華民國 一一五年 六? 月

June 2026

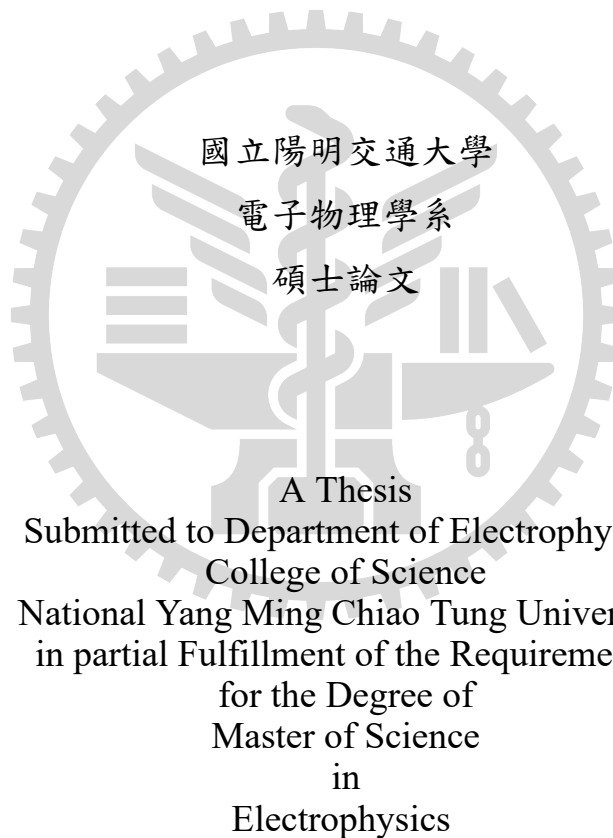
二維材料於純量與向量渦旋光激發下之激子躍遷理論研究
Theoretical studies of excitonic transitions in two dimensional
materials under photoexcitation of scalar and vector vortex
beams

研 究 生：劉人瑩

Student：Liu, Ren-Ying

指導教授：鄭舜仁 博士

Advisor：Dr. Cheng, Shun-Jen



June 2026
Taiwan, Republic of China

中華民國 一 一 五 年 六 月

誌 謝

ACKNOWLEDGEMENT——ACKNOWLEDGEMENT



二維材料於純量與向量渦旋光激發下之激子躍遷理論研究

研 究 生：劉人瑩

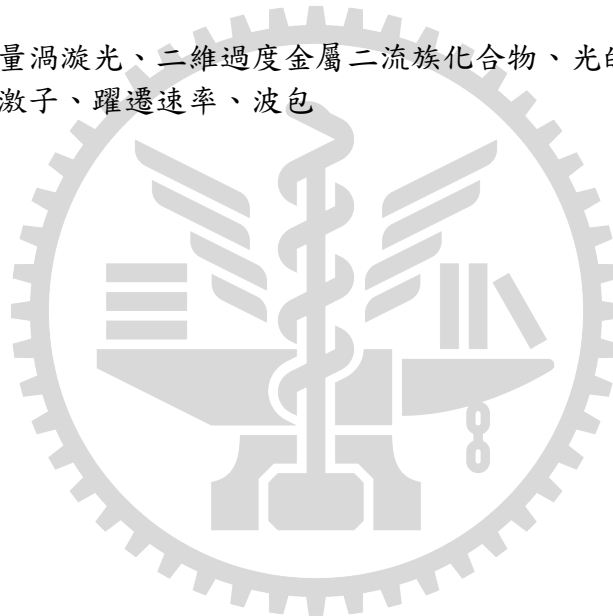
指導教授：鄭舜仁 博士

國立陽明交通大學電子物理學系 碩士班

摘 要

中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要
要中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要中文摘要
中文摘要中文摘要中文摘要中文摘要

關鍵字：純量與向量渦旋光、二維過度金屬二硫族化合物、光的自旋-軌道交互作用、
谷混合亮激子、灰激子、躍遷速率、波包



Contents

Chinese Abstract	i
English Abstract	ii
Contents	iii
Figure Captions	iv
Table Captions	vi
Chapter1 Introduction	1
1.1 Exciton in Transition Metal Dichalcogenide (TMD) Monolayer	1
1.2 Spin and Orbital Angular Momentum (SAM and OAM) of Laguerre–Gaussian (LG) beam	3
1.3 SAM and OAM coupling in LG beam	4
Chapter2 SAM and OAM interaction (SOI) of LG Beam	6
2.1 Gauge Restriction	6
2.1.1 Gauge Freedom in Wave Equations	6
2.1.2 Incompatibility of Gauge Freedom with the Paraxial Approximation	7
2.2 Manifestation of SOI in LG Beam	8
2.2.1 Solutions of LG Beams	9
2.2.2 SOI in the Phase of LG beam’s z Component	11
2.2.3 SOI in the Magnitude of Vector Vortex Beam’s (VVB) z Component	14
Chapter3 Light-Matter Interaction	17
3.1 Theory of Light–Matter Interaction	17
3.1.1 Fermi’s golden rule	17
3.2 Manifestation of SOI in Excitonic Optical Transitions	19
3.2.1 Optical Transition Pattern of Excitons	19
3.2.2 Selective Excitation of Excitons	24
Chapter4 Exciton Wave Packet States	27
4.1 Superposition of Finite-Momentum Exciton (SFME) states	27
4.1.1 Global Phase in Valley-Mixed Exciton	27
4.1.2 Zeroth-Order Approximation	28
4.1.3 Exciton Wave Packet State	30
Chapter5 Summary and outlook	32
References	33
Appendix	36
A.1 Wave equations from Maxwell’s equations in free space	37
A.2 scalar Helmholtz equation	38

Figure Captions

- Figure1.1 (a) Calculated band structures of MoS₂ with different layers, reproduced from Ref. [3]. (b) Side view and (c) top view of transition-metal dichalcogenide (TMD) monolayer with the unit cell framed by the dashed line, reproduced from Ref. [4]. (d) The first Brillouin zone of TMD monolayer in the reciprocal space and the selected high symmetry points 1
- Figure1.2 Illustration of a photon carrying momentum $\mathbf{Q}_c$ exciting (leaving) an electron (hole) with momentum $\mathbf{k} + \mathbf{Q}$ ($-\mathbf{k}$), forming an exciton with center-of-mass (COM) momentum $\mathbf{Q}$. Schematic of ideal of light cone and relation between COM momentum $\mathbf{Q}$ and $\mathbf{Q}_c$ 2
- Figure1.3 (a) Calculated band structure of WSe₂, showing spin-up (red) and spin-down (blue) states, together with a schematic illustration of valley-selective optical transitions at the K and K' valleys, which couple to right- and left-circularly polarized light, respectively, reproduced from Ref. [5]. (b) Schematic configurations of excitonic states, including spin-allowed bright excitons (BXs), momentum-forbidden dark excitons (DXs), and spin-forbidden DXs. 2
- Figure1.4 The lowest valley-degenerate (left) and valley-split (right) exciton bands in MoS₂ along the Q_x direction, reproduced from Ref. [6]. 3
- Figure1.5 Illustration of the radiation patterns of BXs, DXs, and GXs in neutral WSe₂, along with the corresponding optical excitation indicated by arrows (reproduced from Ref. [7]). 3
- Figure1.6 (a) Illustration of the conversion between right-handed and left-handed circularly polarized light generating a torque on a suspended birefringent half-wave plate. (b) Illustration of transforming a Laguerre – Gaussian (LG) mode carrying OAM $\ell\hbar$ per photon into a mode with $-\ell\hbar$ per photon, which produces a mechanical torque on suspended cylindrical lenses. (c) Demonstration of generating an LG beam with nonzero ℓ (left side of bottom row) by introducing a phase shift when converting Hermite–Gaussian modes with zero ℓ (left side of top row). All reproduced from Ref. [10]. 4
- Figure1.7 Illustration of twisted light with a helical wavefront, where each photon carries $\ell\hbar$ of orbital angular momentum. The intensified charge-coupled device camera images show the characteristic doughnut-shaped intensity profile. Reproduced from Ref. [13]. 4
- Figure1.8 Illustration of a beam focused by a lens, showing the variation of SAM before and after the lens along the propagation direction. Reproduced from Ref. [29]. 5
- Figure1.9 (a) Transition-rate distribution for gray excitons in momentum space, reproduced from Ref. [30]. (b) Distribution of the incident light and the excited BX wave-packet states, reproduced from Ref. [6]. 5

Figure2.1	Squared magnitude (intensity) and phase distribution of $u_{\ell,p}(\mathbf{r}_{\parallel})$ at $z = 0$. The green dotted line denotes the beam waist W_0 . The scale bar indicates the length of 1 μm	10
Figure2.2	Squared magnitude and phase distribution of $\mathcal{A}_{\ell,p=0}^L(\mathbf{q}_{\parallel})$ at $z = 0$. The scale bar indicates the considered light-cone range for WSe ₂ -monolayers, $Q_c =$ $8.6 \mu\text{m}^{-1}$	11
Figure2.3	Magnitude (absolute value) of $\mathcal{A}_{q_0}^{\ell,\ell,p=0;C}(\mathbf{r})$ at $z = 0$ for different com- ponents and helicities. The $\sigma\ell$ term leads to asymmetry between cases with the same ℓ in the longitudinal component (a wavelength of 532 nm is assumed). . .	12
Figure2.4	Phase distribution of different components of $\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel})$ for different helic- ities.	13
Figure2.5	The VVB state represented on the Poincaré sphere, parameterized by the azimuthal angle α and polar angle β . Several representative combinations of α and β are illustrated on the sphere surface as examples.	14
Figure2.6	Distribution (squared magnitude) of different components of $\mathcal{A}_{\parallel}^{1,0,-1,0}(\mathbf{q}_{\parallel}; \alpha, \beta)$ on the Poincaré sphere, together with the corresponding polarization states of the transverse components.	16
Figure2.7	Distribution (squared magnitude) of different components of $\mathcal{A}_{\parallel}^{1,-1,-1,-1}(\mathbf{q}_{\parallel}; \alpha, \beta)$ on the Poincaré sphere, together with the corresponding polarization states of the transverse components.	16
Figure3.1	Illustration of the transition dipole moments of valley-mixed BXs [6], and their matching with the transverse polarization of the VVB states $ 1, -1, -1, 1; \alpha =$ $0, \beta = \pi/2\rangle$ and $ 1, -1, -1, 1; \alpha = \pi, \beta = \pi/2\rangle$ in momentum space.	20
Figure3.2	Normalized transition rate of valley-mixed BXs excited by the VVB state $ 1, -1, -1, 1; \alpha, \beta\rangle$ in momentum space.	21
Figure3.3	Normalized transition rate of total BXs excited by the VVB state $ 1, -1, -1, 1; \alpha, \beta\rangle$ in momentum space.	21
Figure3.4	Distribution of the transverse component for the VVB state $ 1, 1, 1, 1; \alpha, \beta\rangle$ in momentum space.	22
Figure3.5	Normalized transition rate of valley-mixed BXs excited by the VVB state $ 1, 1, 1, 1; \alpha, \beta\rangle$ in momentum space.	22
Figure3.6	Normalized transition rate of GXs excited by the VVB state $ 1, -1, -1, 1; \alpha, \beta\rangle$ in momentum space.	23
Figure3.7	Normalized transition rate of GXs excited by the VVB state $ 1, -1, -1, 2; \alpha, \beta\rangle$ in momentum space at the north/south poles of the Poincaré sphere.	23
Figure3.8	Normalized transition rate of GXs excited by the VVB state $ 1, -1, -1, 2; \alpha, \beta\rangle$ in momentum space along the equator of the Poincaré sphere. The scale and color bars are the same as in Figure 3.7.	23
Figure3.9	Normalized total transition rates of BXs, total BXs, and GX excited by a VVB state $ 1, \ell, -1, -\ell; \alpha, \beta = \pi/2\rangle$, for $\ell \in \mathbb{Z}$ with $\ell \in [-5, 5]$, and for azimuthal angles $\alpha = 0, \pi/2, \pi$	25
Figure3.10	Normalized total transition rates of BXs, total BXs, and GXs excited by a VVB state $ 1, -1, -1, 1; \alpha, \beta = \pi/2\rangle$ for azimuthal angles $\alpha = 0$ to 2π	26

Table Captions



Chapter 1 Introduction

1.1 Exciton in Transition Metal Dichalcogenide (TMD) Monolayer

Conventional semiconductors such as diamond, silicon [1], and aluminum arsenide are not suitable for the selective initialization, manipulation, and readout of valley states. This is because they possess multiple inequivalent valleys in their band structures that do not couple distinctly to external optical fields. The situation changed with the discovery of graphene in 2004 [2], which triggered rapid progress in research on two-dimensional (2D) materials. Graphene has a hexagonal honeycomb lattice that places two inequivalent valleys at the K and K' points of the hexagonal Brillouin zone. Nevertheless, the absence of a bandgap in graphene allows electrons to be thermally excited between valleys, making valley polarization unstable at room temperature.

In contrast, transition-metal dichalcogenide (TMD) monolayers share a similar honeycomb lattice but possess a finite direct bandgap (see Figure 1.1) on the order of $\sim 1\text{--}2\text{ eV}$, which lies in the visible to near-infrared spectral range. A TMD monolayer consists of two group-VIA chalcogen atoms, such as sulfur (S) or selenium (Se), with a group-VIB transition-metal atom, such as molybdenum (Mo) or tungsten (W), located between them.

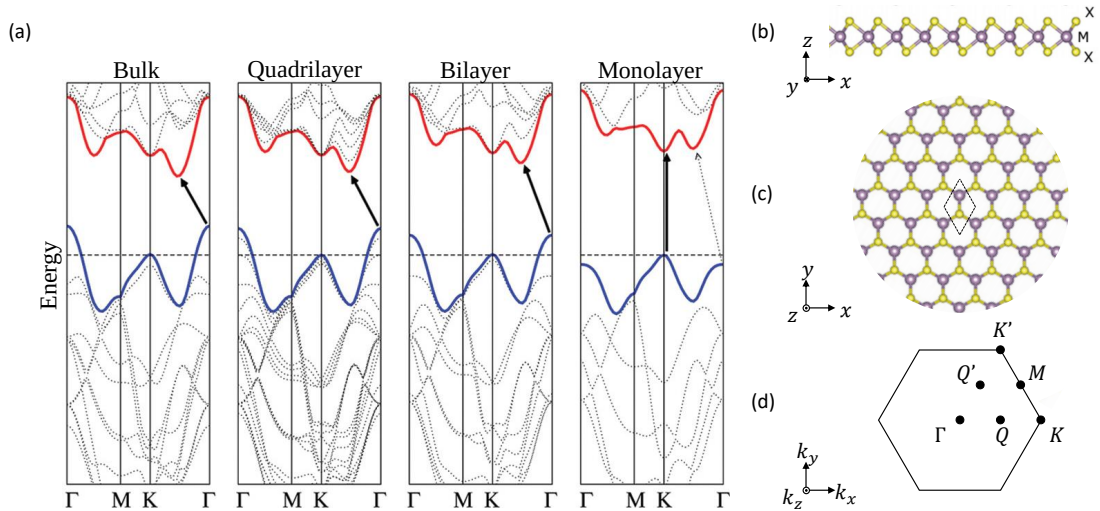


Figure 1.1 : (a) Calculated band structures of MoS₂ with different layers, reproduced from Ref. [3]. (b) Side view and (c) top view of transition-metal dichalcogenide (TMD) monolayer with the unit cell framed by the dashed line, reproduced from Ref. [4]. (d) The first Brillouin zone of TMD monolayer in the reciprocal space and the selected high symmetry points

Under optical excitation, electrons in a TMD monolayer can be promoted from the valence band to the conduction band near the K and K' valleys, leaving holes in the valence band. Due to the strong Coulomb interaction in atomically thin materials with weak dielectric screening, the electron-hole pair can bind together to form a quasiparticle known as an exciton. Figure 1.2 shows that center-of-mass (COM) momentum of exciton is related to momentum of light Q_c with incident angle θ through $Q = Q_c \sin \theta$, which gives the idea of the light cone.

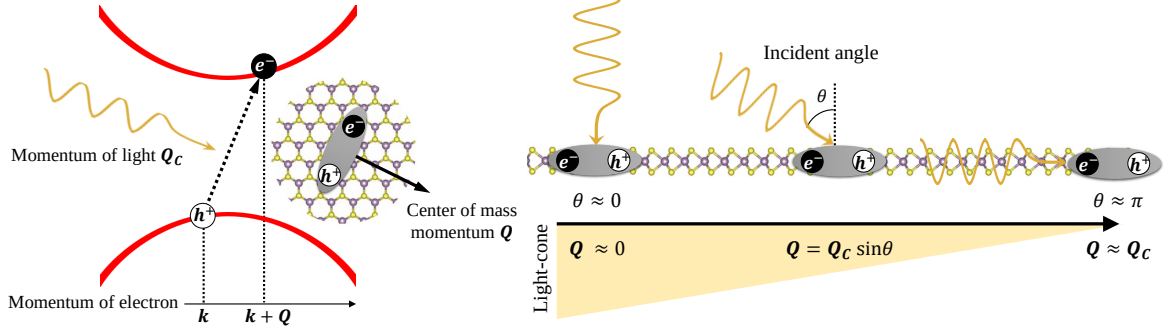


Figure 1.2 : Illustration of a photon carrying momentum Q_c exciting (leaving) an electron (hole) with momentum $k + Q$ ($-k$), forming an exciton with center-of-mass (COM) momentum Q . Schematic of ideal of light cone and relation between COM momentum Q and Q_c

Moreover, the lack of inversion symmetry in the lattice, together with strong spin-orbit coupling from the heavy transition-metal atoms, gives rise to pronounced spin splitting of the electronic bands. At the K/K' valleys, this splitting is typically on the order of a few meV in the conduction band and several hundreds of meV in the valence band, as illustrated in Figure 1.3(a).

The same symmetry considerations also lead to valley-dependent optical selection rules. As shown in Figure 1.3(b), excitons can be classified according to both momentum and spin selection rules. Only excitons with center-of-mass (COM) momentum Q within the light cone, i.e., $|Q| \lesssim Q_c$, and with antiparallel spin configurations can recombine radiatively and are classified as bright excitons (BXs). In contrast, excitons that lie outside the light cone or violate the spin selection rules remain optically inactive and are therefore classified as dark excitons (DXs).

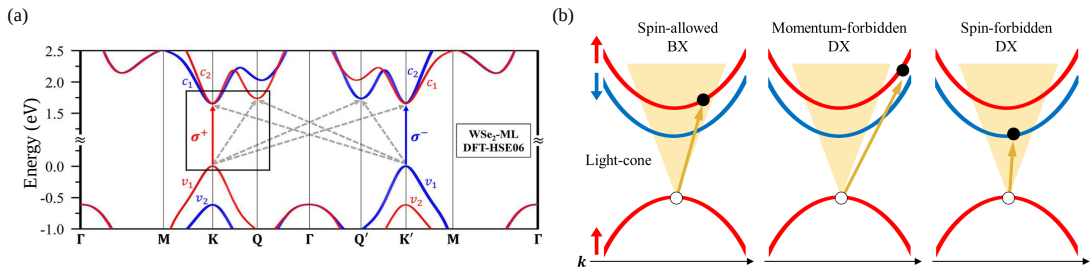


Figure 1.3 : (a) Calculated band structure of WSe_2 , showing spin-up (red) and spin-down (blue) states, together with a schematic illustration of valley-selective optical transitions at the K and K' valleys, which couple to right- and left-circularly polarized light, respectively, reproduced from Ref. [5]. (b) Schematic configurations of excitonic states, including spin-allowed bright excitons (BXs), momentum-forbidden dark excitons (DXs), and spin-forbidden DXs.

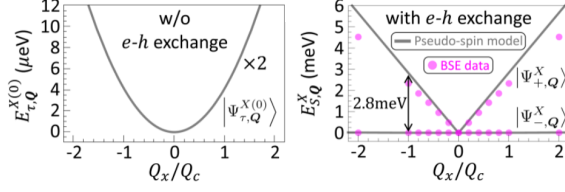


Figure 1.4 : The lowest valley-degenerate (left) and valley-split (right) exciton bands in MoS₂ along the Q_x direction, reproduced from Ref. [6].

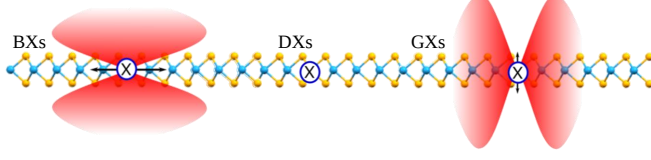


Figure 1.5 : Illustration of the radiation patterns of BXs, DXs, and GXs in neutral WSe₂, along with the corresponding optical excitation indicated by arrows (reproduced from Ref. [7]).

The finer band structure further enriches the classification of quasiparticles. In particular, the lowest bright exciton band, which is initially twofold degenerate, splits into two linearly dispersing branches when electron–hole exchange interactions are included within a pseudospin model. These branches are commonly referred to as valley-mixed excitons [6].

In W-based TMDs, an additional excitonic state known as the gray exciton (GX) can also emerge. As shown in Figure 1.5, GXs emit light polarized perpendicular to the two-dimensional plane (out-of-plane direction). Compared with BXs and DXs, GXs exhibit relatively long lifetimes—typically an order of magnitude longer than BXs ($BX \sim 10$ ps) [7]—while retaining weak but finite optical activity. These properties make GXs particularly useful for studying exciton dynamics and light–matter interactions, and highlight the rich excitonic physics in two-dimensional TMDs.

1.2 Spin and Orbital Angular Momentum (SAM and OAM) of Laguerre–Gaussian (LG) beam

Following the theoretical work of Poynting in 1909 [8] and the experimental demonstration by Beth in 1936 [9], it has been established that each photon in circularly polarized light with finite dimensions [10–12] carries a spin angular momentum (SAM) of $S = \sigma\hbar = \pm\hbar$ along the propagation direction, where σ denotes the photon helicity. In Beth’s experiment, the torque exerted on a suspended half-wave plate by circularly polarized light was measured, demonstrating that spin angular momentum can be transferred to a birefringent medium during propagation (see Figure 1.6(a)).

In 1992, Allen *et al.* showed that a laser beam in a Laguerre–Gaussian (LG) mode possesses a well-defined orbital angular momentum (OAM), also directed along the propagation axis, with $L = \ell\hbar$ per photon, where $\ell \in \mathbb{Z}$ is the azimuthal mode index, commonly referred to as the topological charge [10]. In their experiment, an input Hermite–Gaussian mode, which carries zero OAM, was converted into an LG mode with nonzero OAM. This mode conversion subsequently induced a measurable torque on cylindrical optical elements (see Figure 1.6(b, c)), establishing that light can also carry OAM associated with the spatial structure of its wavefront. The helical phase structure associated with a quantized amount of $\ell\hbar$ of OAM is illustrated

in Figure 1.7, while intensified charge-coupled device (ICCD) images show the characteristic doughnut-shaped intensity distribution.

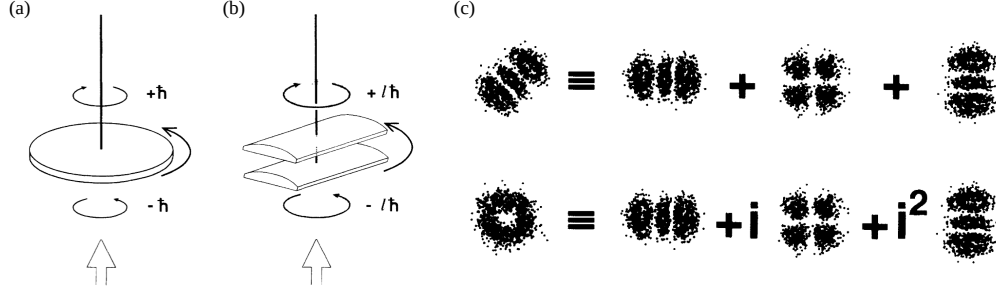


Figure 1.6 : (a) Illustration of the conversion between right-handed and left-handed circularly polarized light generating a torque on a suspended birefringent half-wave plate. (b) Illustration of transforming a Laguerre – Gaussian (LG) mode carrying OAM $\ell\hbar$ per photon into a mode with $-\ell\hbar$ per photon, which produces a mechanical torque on suspended cylindrical lenses. (c) Demonstration of generating an LG beam with nonzero ℓ (left side of bottom row) by introducing a phase shift when converting Hermite–Gaussian modes with zero ℓ (left side of top row). All reproduced from Ref. [10].

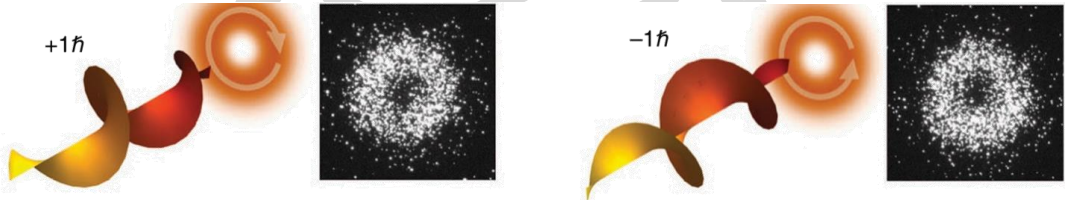


Figure 1.7 : Illustration of twisted light with a helical wavefront, where each photon carries $\ell\hbar$ of orbital angular momentum. The intensified charge-coupled device camera images show the characteristic doughnut-shaped intensity profile. Reproduced from Ref. [13].

Since then, a wide range of applications of light carrying OAM has been developed. Early studies focused on the transfer of angular momentum from light to matter, where objects experience a mechanical torque due to the transfer of OAM, resulting in rotational motion [14–16]. Later developments have extended to various areas, including optical communications [17–19], quantum key distribution [20, 21], astronomy [22], data storage [23, 24], enhanced sensitivity in imaging [25, 26], and optical tweezers [14, 27]. These developments naturally motivate the investigation of how structured light carrying orbital angular momentum interacts with electronic excitations in low-dimensional materials.

1.3 SAM and OAM coupling in LG beam

In free space, coupling between SAM and OAM has been shown to occur in non-paraxial (tightly focused) light. Along the propagation direction, SAM can be partially converted into OAM as the focusing angle increases [28]. This effect can be understood by considering the reduction of the spin component along the beam axis induced by a focusing lens (see Figure 1.8).

The initial spin s_0 is aligned with the propagation direction. After focusing, however, the spin is no longer parallel to the beam axis; instead, the final spin s_1 is tilted, such that only its longitudinal component s_z contributes to the total spin angular momentum of the beam. Since the maximum magnitude of the spin angular momentum is $s\hbar$ per photon, the total spin after focusing must be reduced. However, the total angular momentum is conserved during the focusing process. Consequently, the reduction in SAM must be compensated by the generation of orbital angular momentum, leading to a transfer from SAM to OAM [29].

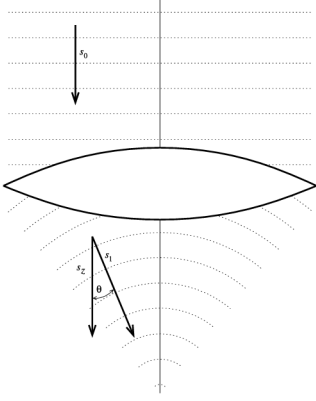


Figure 1.8 : Illustration of a beam focused by a lens, showing the variation of SAM before and after the lens along the propagation direction. Reproduced from Ref. [29].

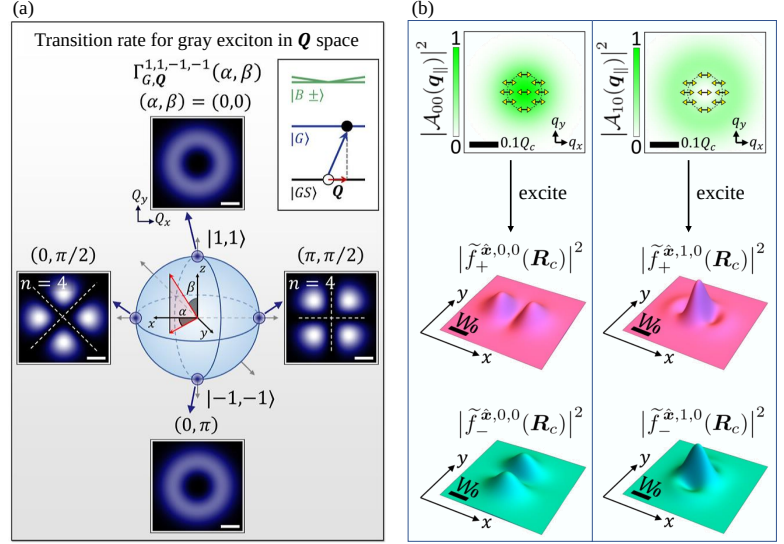


Figure 1.9 : (a) Transition-rate distribution for gray excitons in momentum space, reproduced from Ref. [30]. (b) Distribution of the incident light and the excited BX wave-packet states, reproduced from Ref. [6].

However, it should be emphasized that the simple proportionality between spin and orbital angular momentum with the optical helicity σ and topological charge ℓ holds strictly only within the paraxial approximation [31, 32], which often neglects the longitudinal component along the propagation direction. However, it is precisely this longitudinal component that can couple to GXs. More importantly, this neglected component already contains two distinct types of spin-orbit interaction (SOI).

In 2024, Oscar Javier Gomez Sanchez *et al.* demonstrated that a type of SOI originating from the phase of the longitudinal component induces momentum-dependent transition rates of GXs through the superposition of two twisted-light beams, forming a vector vortex beam (VVB) (see Figure 1.9(a)) [30]. However, the combined contribution of SAM and OAM was not fully clarified in their formulation. Meanwhile, in 2022, Guan-Hao Peng *et al.* investigated the photoexcitation of valley-mixed BXs by LG beams within the pseudospin model, suggesting a connection to excitonic topology [6] (see Figure 1.9(b)). Nevertheless, the global phase of the valley-mixed excitons may still influence the excited wave-packet profile.

In this work, we extend the studies of Oscar Javier Gomez Sanchez *et al.* and Guan-Hao Peng *et al.* to investigate the interplay between spin and orbital angular momentum of light in determining transition rates and wave-packet distributions of excitonic states within SI units.

Chapter 2 SAM and OAM interaction (SOI) of LG Beam

2.1 Gauge Restriction

The derivation of Laguerre–Gaussian (LG) beams involves a series of approximations, among which the paraxial approximation plays a central role. Importantly, the wave equation obtained under the paraxial approximation is not independent of the choice of gauge; rather, it is implicitly constrained by the adopted gauge condition.

In this section, we examine how gauge choice restricts the formulation of wave equations. We begin by introducing the wave equations under different gauge conditions, and then analyze the compatibility of the paraxial approximation. This discussion clarifies why certain gauges are more suitable than others for describing paraxial optical fields.

2.1.1 Gauge Freedom in Wave Equations

Maxwell's equations describe how the **electric field** $\mathbf{E}(\mathbf{r}, t)$ and the **magnetic field** $\mathbf{B}(\mathbf{r}, t)$ evolve in space and time. These equations form the fundamental basis for describing electromagnetic waves, including Laguerre–Gaussian beams. In **free space**, the charge density ρ and current density $\mathbf{J}$ vanish. In this case, Maxwell's equations reduce to four homogeneous differential equations involving the vacuum permittivity ϵ_0 , the vacuum permeability μ_0 , and the speed of light in vacuum $c = 1/\sqrt{\mu_0\epsilon_0}$.

Within this electrodynamics framework, Maxwell's equations can be rewritten as two coupled wave equations for the **scalar and vector potentials**, $\phi(\mathbf{r}, t)$ and $\mathbf{A}(\mathbf{r}, t)$, derived in A.1 as

$$\nabla^2 \phi(\mathbf{r}, t) + \frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}(\mathbf{r}, t)) = 0, \quad (2.1)$$

$$\nabla^2 \mathbf{A}(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}(\mathbf{r}, t) - \nabla \left(\nabla \cdot \mathbf{A}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial}{\partial t} \phi(\mathbf{r}, t) \right) = 0. \quad (2.2)$$

These equations contain the same physical information as the original Maxwell equations, but are expressed in terms of the scalar and vector potentials.

Further simplification of Eqs. (2.1) and (2.2) can be achieved by exploiting the gauge freedom of the potentials. Below, we introduce two commonly used gauges: the **Coulomb gauge** and the **Lorenz gauge**. Superscripts C and L are used to distinguish quantities defined in the

Coulomb and Lorenz gauges, respectively. The gauge conditions are given by

$$\text{Coulomb gauge condition : } \nabla \cdot \mathbf{A}^C(\mathbf{r}, t) = 0, \quad (2.3)$$

$$\text{Lorenz gauge condition : } \nabla \cdot \mathbf{A}^L(\mathbf{r}, t) = -\frac{1}{c^2} \frac{\partial}{\partial t} \phi^L(\mathbf{r}, t). \quad (2.4)$$

Both gauges simplify the wave equations. In the Coulomb gauge, they reduce to

$$\nabla^2 \phi^C(\mathbf{r}, t) = 0, \text{ and} \quad (2.5)$$

$$\nabla^2 \mathbf{A}^C(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^C(\mathbf{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \nabla \phi^C(\mathbf{r}, t) = 0, \quad (2.6)$$

where the last equality follows from Eq. (2.5). In the Lorenz gauge, Eqs. (2.1) and (2.2) become

$$\nabla^2 \phi^L(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \phi^L(\mathbf{r}, t) = 0, \text{ and} \quad (2.7)$$

$$\nabla^2 \mathbf{A}^L(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}^L(\mathbf{r}, t) = 0. \quad (2.8)$$

The physical field $\mathbf{B}(\mathbf{r}, t)$ and $\mathbf{E}(\mathbf{r}, t)$ can be related to the vector potentials defined in different gauges through Eq. (A.1.5) and

$$\mathbf{E}(\mathbf{r}) = i\omega \mathbf{A}^C(\mathbf{r}) = i\omega \left(\mathbf{A}^L(\mathbf{r}) + \frac{\nabla (\nabla \cdot \mathbf{A}^L(\mathbf{r}))}{q_0^2} \right). \quad (2.9)$$

The derivation of this relation assumes a **monochromatic wave** ansatz. Specifically, the fields and potentials are written as $\mathbf{E}(\mathbf{r}, t) = \text{Re} [\mathbf{E}(\mathbf{r})e^{-i\omega t}]$, $\mathbf{A}^C(\mathbf{r}, t) = \text{Re} [\mathbf{A}^C(\mathbf{r})e^{-i\omega t}]$, $\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r})e^{-i\omega t}]$, and $\phi^L(\mathbf{r}, t) = \text{Re} [\phi^L(\mathbf{r})e^{-i\omega t}]$, where ω is the angular frequency of light. Substituting these expressions into Eq. (A.1.7) and applying the corresponding gauge conditions leads to Eq. (2.9). Here $q_0 = \omega/c$ denotes the wavenumber of light in free space.

2.1.2 Incompatibility of Gauge Freedom with the Paraxial Approximation

In principle, the vector potential formulation of LG beams can be derived starting from either Eq. (2.6) or Eq. (2.8). However, only the Lorenz gauge remains fully consistent throughout the derivation, whereas the Coulomb gauge condition becomes incompatible with the paraxial wave ansatz. For this reason, we adopt the Lorenz gauge in the following derivation for simplicity and consistency, and verify this choice at the end.

Before applying the paraxial approximation, the scalar Helmholtz equation is obtained from Eq. (2.8) by imposing both the monochromatic ($\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r})e^{-i\omega t}]$) and transverse wave ansätze ($\mathbf{A}^L(\mathbf{r}) = \hat{\varepsilon} A^L(\mathbf{r})$, where $\hat{\varepsilon}$ is a polarization unit vector) [33] within the Lorenz gauge (see Sec. A.2):

$$\nabla^2 A^L(\mathbf{r}) + q_0^2 A^L(\mathbf{r}) = 0. \quad (2.10)$$

The **paraxial approximation** [34–36] assumes that the rays have only small inclinations with respect to the optical axis in geometric optics. In wave optics, this corresponds to an angular

spectrum dominated by plane-wave components whose wavevectors are close to the propagation direction. Here, we choose the propagation direction along the z axis, such that the dominant component of the wavevector $\mathbf{q}_0$ lies along $\hat{z}$. Under this approximation, we introduce the **paraxial wave** ansatz, $A^L(\mathbf{r}) = U(\mathbf{r})e^{iq_0z}$, and substitute it into Eq. (2.10), which yields

$$\nabla_{\parallel}^2 U(\mathbf{r}) + \frac{\partial^2}{\partial z^2} U(\mathbf{r}) + 2iq_0 \frac{\partial}{\partial z} U(\mathbf{r}) = 0. \quad (2.11)$$

Here, $\nabla_{\parallel}^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ denotes the transverse Laplacian. The paraxial wave ansatz assumes that the **envelope function** $U(\mathbf{r})$ **varies slowly along the z direction** compared to the rapid phase variation of e^{iq_0z} . Consequently, the paraxial approximation implies the following relations:

$$\left| \frac{\partial^2}{\partial z^2} U(\mathbf{r}) \right| \ll |\nabla_{\parallel}^2 U(\mathbf{r})|, \quad \left| \frac{\partial^2}{\partial z^2} U(\mathbf{r}) \right| \ll q_0 \left| \frac{\partial}{\partial z} U(\mathbf{r}) \right|.$$

Neglecting the second-order derivative with respect to z then leads to the paraxial Helmholtz equation

$$\nabla_{\parallel}^2 U(\mathbf{r}) + 2iq_0 \frac{\partial}{\partial z} U(\mathbf{r}) = 0. \quad (2.12)$$

We now clarify why the Coulomb gauge is not suitable in this derivation. The paraxial wave ansatz generally does not satisfy the Coulomb gauge condition (Eq. (2.3)), which requires

$$\nabla \cdot \mathbf{A}^C(\mathbf{r}, t) = \hat{\epsilon} \cdot \nabla A^C(\mathbf{r}, t) = 0. \quad (2.13)$$

For a transverse wave with the polarization unit vector $\hat{\epsilon}$ confined to the x - y plane [33], this condition reduces to

$$\hat{\epsilon} \cdot \nabla_{\parallel} A^C(\mathbf{r}, t) = 0, \quad (2.14)$$

where $\nabla_{\parallel} = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y}$. This condition constrains the transverse spatial variation of $A^C(\mathbf{r}, t)$ along the polarization direction, which is incompatible with the assumed slowly varying envelope function $U(\mathbf{r})$ [34]. In other words, the Coulomb gauge imposes an additional constraint that conflicts with the paraxial wave ansatz. To satisfy the divergence-free condition, a nonzero longitudinal (z) component of the vector potential must be introduced, going beyond the simple transverse ansatz. Therefore, the Lorenz gauge provides a more natural and consistent framework for deriving the paraxial Helmholtz equation and LG beam solutions.

2.2 Manifestation of SOI in LG Beam

Within the paraxial approximation, the optical helicity σ and the topological charge ℓ are directly proportional to the spin and orbital angular momentum of light, respectively [31, 32]. This correspondence provides a convenient framework for analyzing the angular momentum properties of LG beams.

Moreover, when the vector potential formulation of LG beams is expressed in the Coulomb gauge, an SOI can emerge through the additional longitudinal component of the field, giving

rise to terms proportional to $\sigma\ell$ and $\sigma + \ell$. Adopting the Coulomb gauge is further justified by the theory of light-matter interaction, where the vector potential is conventionally formulated within both the Coulomb gauge and the angular spectrum representation. In this representation, approximately interpret the in-plane wavevector as a vector in reciprocal space enable a connection to theory of matter.

In the following section, we first introduce the LG beam solutions in both real-space and angular spectrum representations. We then perform a gauge transformation to reveal the SOI.

2.2.1 Solutions of LG Beams

Under the **long Rayleigh-length range** [37], the LG beam in real space can be solved from Eq. (2.12) within cylindrical coordinates as

$$\mathbf{A}_{\mathbf{q}_0}^{\hat{\epsilon}, \ell, p; L}(\mathbf{r}) = \hat{\epsilon} A_0^{\text{LG}} u_{\ell p}(\mathbf{r}_{\parallel}) e^{iq_0 z}, \quad (2.15)$$

where the position vector is written as $\mathbf{r} = (\rho, \phi, z) = (\mathbf{r}_{\parallel}, z)$. Here $\mathbf{r}_{\parallel}$ denotes the in-plane position vector consisting of the radial coordinate ρ and the azimuthal angle ϕ . The integer $\ell \in \mathbb{Z}$ is the **topological charge**, and the integer $p \geq 0$ is the radial mode index. The vector $\mathbf{q}_0 = q_0 \hat{z}$ denotes the wavevector along the propagation direction, and A_0^{LG} is a constant amplitude. The envelope function is expressed as

$$U(\rho, \phi, z) \approx u_{\ell p}(\mathbf{r}_{\parallel}) = f_{\ell p}(\rho) e^{i\ell\phi}, \quad (2.16)$$

where the radial function is given by [6, 30, 37] $f_{\ell p}(\rho) = \sqrt{\frac{2p!}{\pi(|\ell|+p)!}} L_p^{|\ell|} \left(\frac{2\rho^2}{W_0^2} \right) \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} \exp\left(-\frac{\rho^2}{W_0^2}\right)$. Here the generalized Laguerre polynomial is defined as $L_p^{|\ell|}(x) = \sum_{m=0}^p (-1)^m \frac{(|\ell|+p)!}{(p-m)! (|\ell|+m)! m!} x^m$, with $x = \frac{2\rho^2}{W_0^2}$. The parameter W_0 denotes the beam waist of the LG beam, while the cylindrical coordinates are related to Cartesian coordinates through $\rho = \sqrt{x^2 + y^2}$ and $\phi = \tan^{-1}(y/x)$.

Focusing on the minimum cross-section area of the beam ($z = 0$), Figure 2.1 shows that as ℓ and/or p increase, $u_{\ell p}(\mathbf{r}_{\parallel})$ develops a broader doughnut-shaped intensity profile, accompanied by an increasing number of radial nodes along the radial direction. The phase varies continuously with the azimuthal angle ϕ , reflecting the helical wavefront associated with orbital angular momentum. In particular, the phase winds by $2\pi\ell$ around the beam center. The beam waist is chosen as $W_0 = 1.5 \mu\text{m}$, which sets the spatial scale of the beam and determines the radial extent of the intensity distribution.

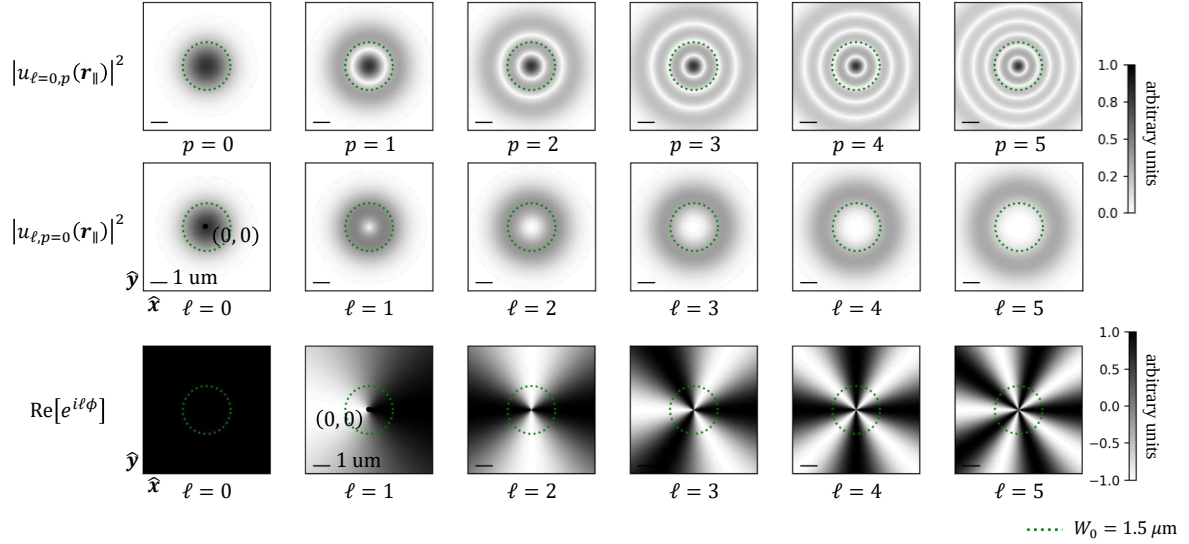


Figure 2.1 : Squared magnitude (intensity) and phase distribution of $u_{\ell,p}(\mathbf{r}_{\parallel})$ at $z = 0$. The green dotted line denotes the beam waist W_0 . The scale bar indicates the length of 1 μm .

To derive the vector potential of LG beams in the angular spectrum representation, the field is expanded in a plane-wave basis as

$$\mathbf{A}(\mathbf{r}) = \sum_{\mathbf{q}} \mathcal{A}(\mathbf{q}) e^{i\mathbf{q} \cdot \mathbf{r}}, \quad (2.17)$$

where $\mathcal{A}(\mathbf{q})$ is the complex amplitude associated with each plane-wave component $e^{i\mathbf{q} \cdot \mathbf{r}}$. The LG beams in Eq. (2.15) can be written in this form through a defined **two-dimensional Fourier transformation** and its corresponding inverse transformation,

$$u_{\ell p}(\mathbf{r}_{\parallel}) = \frac{1}{(2\pi)^2} \int d^2 \mathbf{q}_{\parallel} F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}}, \quad \longleftrightarrow \quad F_{\ell p}^{\text{LG}}(\mathbf{q}_{\parallel}) = \int d^2 \mathbf{r}_{\parallel} u_{\ell p}(\mathbf{r}_{\parallel}) e^{-i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}}. \quad (2.18)$$

For applications to two-dimensional materials, the in-plane wavevector $\mathbf{q}_{\parallel}$ can be approximately regarded as a vector in reciprocal space. Under this approximation, Eq. (2.18) can be discretized, and the Eq. (2.17) becomes

$$\mathbf{A}_{\mathbf{q}_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}) = \hat{\varepsilon} \sum_{\mathbf{q}_{\parallel}} \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{iq_0 z}. \quad (2.19)$$

Here the in-plane wavevector is written as $\mathbf{q}_{\parallel} = (q_{\parallel}, \phi_{\mathbf{q}_{\parallel}})$. Since the spatial distribution has the form $u_{\ell p}(\mathbf{r}_{\parallel}) = f_{\ell p}(\rho) e^{i\ell\phi}$, the complex amplitude can be expressed analytically as

$$\mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) = \tilde{F}_{\ell p}(q_{\parallel}) e^{i\ell\phi_{\mathbf{q}_{\parallel}}}, \quad (2.20)$$

where $\tilde{F}_{\ell p}(q_{\parallel}) = \frac{A_0^{\text{LG}}}{\Omega} (-1)^{|\ell|} 2\pi \mathcal{H}_{|\ell|}[f_{\ell p}(\rho)]$, and $\mathcal{H}_{|\ell|}[f_{\ell p}(\rho)] = \int_0^\infty d\rho \rho f_{\ell p}(\rho) J_{|\ell|}(q_{\parallel} \rho)$ is the **Hankel transform** of order $|\ell|$, with $J_{|\ell|}(q_{\parallel} \rho)$ being the **Bessel function of the first kind**. Here Ω denotes the area of the two-dimensional system.

In this study, we focus on the fundamental radial mode $p = 0$. In real space, the vector

potentials take the explicit form

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; L}(\mathbf{r}) = \hat{\varepsilon} \frac{A_0^{\text{LG}}}{\sqrt{|\ell|!} \pi/2} \left(\frac{\sqrt{2}\rho}{W_0} \right)^{|\ell|} \exp\left(-\frac{\rho^2}{W_0^2}\right) \exp[i(\ell\phi + q_0 z)], \quad (2.21)$$

whereas in the angular spectrum representation the amplitude become

$$\mathcal{A}_{\ell, p=0}^L(\mathbf{q}_{\parallel}) = \left(\frac{\pi A_0^{\text{LG}} W_0^2 (-1)^{|\ell|}}{\Omega \sqrt{|\ell|!} \pi/2} \right) \left[\left(\frac{q_{\parallel} W_0}{\sqrt{2}} \right)^{|\ell|} \exp\left(-\frac{q_{\parallel}^2 W_0^2}{4}\right) \right] \exp(i\ell\phi_{q_{\parallel}}). \quad (2.22)$$

Similarly, focusing on the $z = 0$ plane, Figure 2.2 shows that $\mathcal{A}_{\ell, p=0}^L(\mathbf{q}_{\parallel})$ exhibits an intensity profile analogous to that of $u_{\ell, p=0}(\mathbf{r}_{\parallel})$, reflecting the characteristic doughnut-like structure in angular spectrum. This correspondence follows directly from the Fourier transform relation between two representations of the LG beam. The doughnut-shaped distribution in angular spectrum indicates that the dominant in-plane wavevectors lie within a finite range determined by the beam waist W_0 .

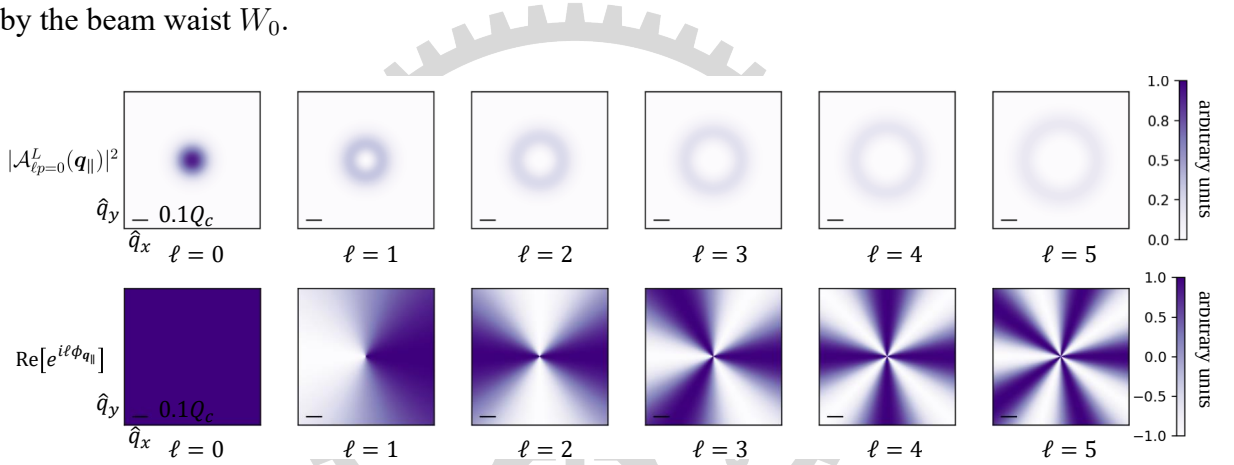


Figure 2.2 : Squared magnitude and phase distribution of $\mathcal{A}_{\ell, p=0}^L(\mathbf{q}_{\parallel})$ at $z = 0$. The scale bar indicates the considered light-cone range for WSe₂-monolayers, $Q_c = 8.6 \mu\text{m}^{-1}$.

2.2.2 SOI in the Phase of LG beam's z Component

In the previous section, Eq. (2.9) was introduced to relate the vector potentials defined in the two gauges to the physical fields. Using this relation, we can perform gauge transformation by rewritten Eq. (2.9) as

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; C}(\mathbf{r}) = \mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}) + \frac{\nabla (\nabla \cdot \mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p; L}(\mathbf{r}))}{q_0^2}. \quad (2.23)$$

To explicitly incorporate the SAM of light, the polarization unit vector is defined as $\hat{\varepsilon} = \frac{1}{\sqrt{2}} (\hat{x} + i\sigma\hat{y})$, where the photon helicity $\sigma = \pm 1$ denotes right- and left-handed circular polarizations. For LG beams in the fundamental radial mode $p = 0$, the vector potential in real space under the

Coulomb gauge is obtained as

$$\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r}) \approx A_0^{\text{LG}} \left[f_{\ell, p=0}(\rho) e^{i\ell\phi} \hat{\varepsilon} + \frac{i}{\sqrt{2}q_0\rho} \left((|\ell| + \sigma\ell) - \frac{2\rho^2}{W_0^2} \right) f_{\ell, p=0}(\rho) e^{i(\sigma+\ell)\phi} \hat{z} \right] e^{iq_0 z}. \quad (2.24)$$

In general, the second term in Eq. (2.23) introduces corrections to all spatial components of the vector potential. However, under the paraxial approximation, these corrections are suppressed. Therefore, we retain only the longitudinal (z) component as the leading correction.

The longitudinal component in Eq. (2.24) reveals two distinct types of SOI: a $\sigma\ell$ dependence in the amplitude and a $\sigma + \ell$ dependence in the phase. Figure 2.3 shows that the magnitude (absolute value) at $z = 0$ of this longitudinal component is significantly weaker than that of the transverse components. The $\sigma\ell$ dependence manifests as an asymmetry between cases with the same ℓ but opposite helicities σ .

It is worth noting that the longitudinal term is weighted by a factor proportional to $(q_0 W_0^2)^{-1}$, which is inversely related to the Rayleigh range ($\frac{1}{2}q_0 W_0^2$) [35]. This indicates that a smaller beam waist W_0 leads to a stronger longitudinal component, corresponding to a shorter Rayleigh range and tighter focusing, as shown in the bottom row of Figure 2.3. The beam waist W_0 is chosen to be $1.5 \mu\text{m}$ and $0.5 \mu\text{m}$, representing a typical paraxial beam and a case at the boundary of the paraxial regime, respectively [38–40]. This demonstrates that the longitudinal component, although small, remains finite even within the paraxial regime.

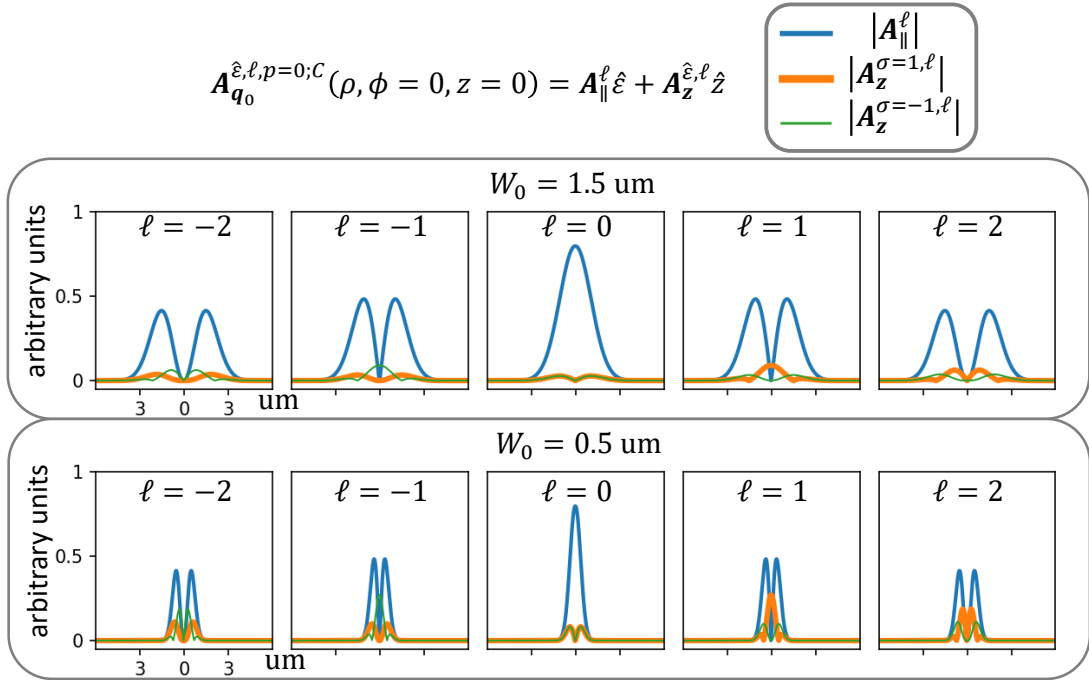


Figure 2.3 : Magnitude (absolute value) of $\mathbf{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r})$ at $z = 0$ for different components and helicities. The $\sigma\ell$ term leads to asymmetry between cases with the same ℓ in the longitudinal component (a wavelength of 532 nm is assumed).

As for the $\sigma + \ell$ dependence in the phase term, it is retained in the angular spectrum representation and will be discussed below. By combining Eq. (2.23) with the two-dimensional

Fourier transformation defined in Coulomb gauge, vector potential can be written as

$$\mathcal{A}_{q_0}^{\hat{\varepsilon}, \ell, p=0; C}(\mathbf{r}) = \frac{1}{(2\pi)^2} \int d^2 \mathbf{q}_{\parallel} \mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) e^{i \mathbf{q}_{\parallel} \cdot \mathbf{r}_{\parallel}} e^{i q_0 z}. \quad (2.25)$$

The corresponding complex amplitude in the angular spectrum representation is obtained as

$$\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) = \left[\hat{\varepsilon} - \left(\frac{|\mathbf{q}|}{q_0} \right)^2 (\hat{\varepsilon} \cdot \hat{\mathbf{q}}_{\parallel}) \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.26)$$

$$= \left[\hat{\varepsilon} - \left(\frac{|\mathbf{q}|}{q_0} \right)^2 \frac{\sin \theta_q}{\sqrt{2}} e^{i \sigma \phi_{q_{\parallel}}} \hat{\mathbf{q}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}) \quad (2.27)$$

$$\approx \left[\hat{\varepsilon} - \frac{\sin \theta_q}{\sqrt{2}} e^{i \sigma \phi_{q_{\parallel}}} \hat{\mathbf{z}} \right] \mathcal{A}_{\ell p}^L(\mathbf{q}_{\parallel}), \quad (2.28)$$

where the last approximation follows from the paraxial condition $\frac{q_0}{|\mathbf{q}|} = \cos \theta_q \approx 1$. The unit vector is given by $\hat{\mathbf{q}} = \sin \theta_q (\cos \phi_{q_{\parallel}} \hat{\mathbf{x}} + \sin \phi_{q_{\parallel}} \hat{\mathbf{y}}) + \cos \theta_q \hat{\mathbf{z}} \approx \hat{\mathbf{z}}$, and $\sin \theta_q = \frac{q_{\parallel}}{\sqrt{q_{\parallel}^2 + q_0^2}}$. Substituting the Lorenz gauge amplitude obtained previously, the Coulomb gauge complex amplitude in the angular spectrum representation becomes

$$\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel}) \approx \tilde{F}_{\ell p}(q_{\parallel}) e^{i \ell \phi_{q_{\parallel}}} \hat{\varepsilon} - \frac{\sin \theta_q}{\sqrt{2}} \tilde{F}_{\ell p}(q_{\parallel}) e^{i(\sigma + \ell) \phi_{q_{\parallel}}} \hat{\mathbf{z}}. \quad (2.29)$$

Notably, the SOI remains encoded in the phase of the longitudinal component through the $\sigma + \ell$ dependence, corresponding to the total angular momentum (TAM), as illustrated in Figure 2.4. However, this phase contribution is generally difficult to detect experimentally, as it tends to average out in intensity-based measurements. To access this TAM contribution, one can employ the interference of two LG beams, forming so-called vector vortex beams (VVBs), which will be discussed in the following section.

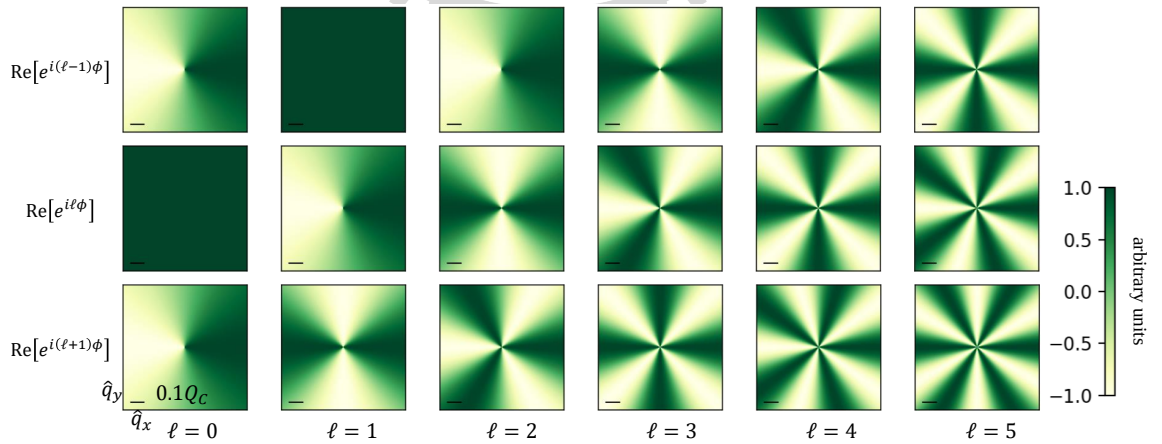


Figure 2.4 : Phase distribution of different components of $\mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel})$ for different helicities.

2.2.3 SOI in the Magnitude of Vector Vortex Beam's (VVB) z Component

In Ref. [30], Oscar Javier Gomez Sanchez *et al.* pointed out that the information of total angular momentum (TAM) encoded in the phase can be lost when calculating optical transition rates. To retain and control this information, they introduced vector vortex beams (VVBs), which are constructed by superimposing two twisted light beams. The VVB state is expressed as a superposition through

$$\mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) = \cos\left(\frac{\beta}{2}\right) \mathcal{A}^{\sigma,\ell}(\mathbf{q}_{\parallel}) + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) \mathcal{A}^{\sigma',\ell'}(\mathbf{q}_{\parallel}). \quad (2.30)$$

Here, the vector potential $\mathcal{A}^{\sigma,\ell}(\mathbf{q}_{\parallel}) = \mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel})$, the complex amplitude in Eq. (2.29), represents a LG beam characterized by SAM (σ) and OAM (ℓ). In this study, the radial mode index is set to $p = 0$, as the primary focus remains on the interplay between SAM and OAM. The second term in the expression corresponds to another LG beam carrying a different combination of angular momentum (σ', ℓ').

The parameter α controls the relative phase between the two beams, while β determines their relative weight in the superposition. These two parameters can be interpreted as the azimuthal and polar angles on the Poincaré sphere (see Figure 2.5), providing a convenient geometric representation of the VVB state, which can be written as

$$|\sigma, \ell, \sigma', \ell'; \alpha, \beta\rangle = \cos\left(\frac{\beta}{2}\right) |\sigma, \ell\rangle + e^{i\alpha} \sin\left(\frac{\beta}{2}\right) |\sigma', \ell'\rangle. \quad (2.31)$$

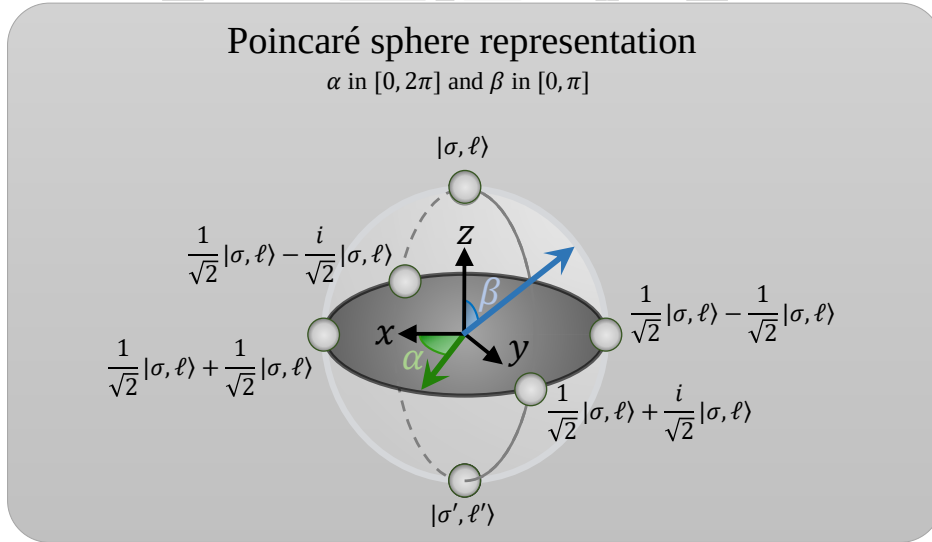


Figure 2.5 : The VVB state represented on the Poincaré sphere, parameterized by the azimuthal angle α and polar angle β . Several representative combinations of α and β are illustrated on the sphere surface as examples.

Folowing Eq. (2.29), a VVB state also contains both transverse and longitudinal compo-

nents, which can be written as

$$\mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) = \mathcal{A}_{\parallel}^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \hat{\mathbf{e}} + \mathcal{A}_z^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \hat{\mathbf{z}}. \quad (2.32)$$

The squared magnitudes of the transverse and longitudinal components are obtained as

$$\begin{aligned} \left| \mathcal{A}_{\parallel}^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2 &= \cos^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell_p}(q_{\parallel}) \right|^2 + \sin^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell'_p}(q_{\parallel}) \right|^2 \\ &\quad + \frac{1 + \sigma\sigma'}{2} \sin \beta \tilde{F}_{\ell_p}(q_{\parallel}) \tilde{F}_{\ell'_p}^*(q_{\parallel}) \cos\left(\Delta \ell \phi_{\mathbf{q}_{\parallel}} + \alpha\right), \end{aligned} \quad (2.33)$$

$$\begin{aligned} \left| \mathcal{A}_z^{\sigma,\ell,\sigma',\ell'}(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2 &= \frac{\sin^2 \theta_{\mathbf{q}}}{2} \left[\cos^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell_p}(q_{\parallel}) \right|^2 + \sin^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell'_p}(q_{\parallel}) \right|^2 \right. \\ &\quad \left. + \sin \beta \tilde{F}_{\ell_p}(q_{\parallel}) \tilde{F}_{\ell'_p}^*(q_{\parallel}) \cos\left(\Delta J \phi_{\mathbf{q}_{\parallel}} + \alpha\right) \right], \end{aligned} \quad (2.34)$$

where the last term in each expression represents the interference between the two LG beams. For the transverse component, the interference vanishes when the two beams carry opposite SAM ($\sigma' = -\sigma$), since in that case $1 + \sigma\sigma' = 0$. Here,

$$\Delta \ell = \ell' - \ell \quad (2.35)$$

is the difference in OAM between the two beams, while

$$\Delta J = (\ell' + \sigma') - (\ell + \sigma) \quad (2.36)$$

is the difference in TAM. In contrast to the transverse component, the longitudinal component explicitly contains the combined SAM–OAM contribution through ΔJ , reflecting its origin from the longitudinal field and encoding this structure in the amplitude rather than in the phase. In addition, the conditions for the disappearance of the azimuthal dependence on $\phi_{\mathbf{q}_{\parallel}}$ are different for the two components. For the transverse component, the azimuthal dependence vanishes when $\ell' = \ell$, whereas for the longitudinal component it disappears when $\Delta J = 0$.

For simplicity, we consider a VVB composed of two LG beams with opposite SAM and the same OAM absolute value ($\sigma' = -\sigma$ and $|\ell'| = |\ell|$). The components then reduce to

$$\left| \mathcal{A}_{\parallel}^{\sigma,\ell,-\sigma,\ell}(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2 = \left| \tilde{F}_{\ell_p}(q_{\parallel}) \right|^2, \quad (2.37)$$

$$\left| \mathcal{A}_z^{\sigma,\ell,-\sigma,\ell}(\mathbf{q}_{\parallel}; \alpha, \beta) \right|^2 = \frac{\sin^2 \theta_{\mathbf{q}}}{2} \left| \tilde{F}_{\ell_p}(q_{\parallel}) \right|^2 \left[1 + \sin \beta \cos\left(\Delta J \phi_{\mathbf{q}_{\parallel}} + \alpha\right) \right]. \quad (2.38)$$

In this case, $\Delta J = -2\sigma$. The transverse component therefore exhibits an isotropic squared magnitude in angular spectrum and does not retain information about the SOI. By contrast, the longitudinal component reflects the TAM difference between the two LG beams through the interference term. Here, both beams share the same radial factor $\tilde{F}_{\ell_p}(q_{\parallel})$ for $|\ell'| = |\ell|$. As a result, the common radial factor can be factored out, allowing us to focus directly on the angular interference.

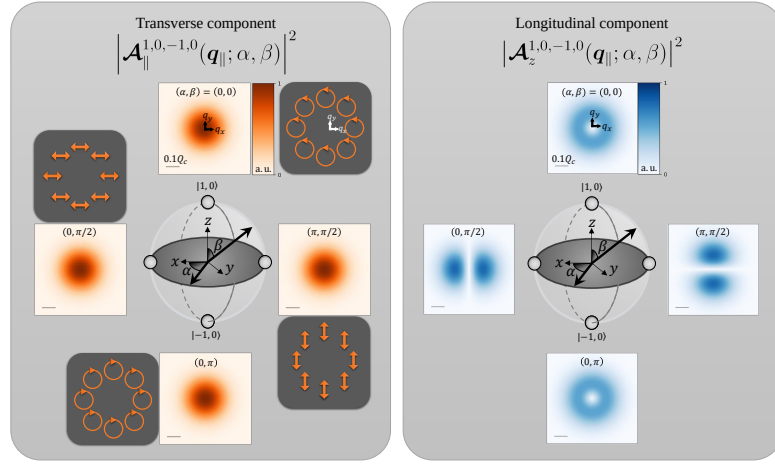


Figure 2.6 : Distribution (squared magnitude) of different components of $\mathcal{A}_{\parallel}^{1,0,-1,0}(\mathbf{q}_{\parallel}; \alpha, \beta)$ on the Poincaré sphere, together with the corresponding polarization states of the transverse components.

For a simple case $\sigma' = -\sigma = -1$ and $\ell' = \ell = 0$, Figure 2.6 shows that the squared magnitude of the longitudinal component exhibits TAM dependent azimuthal modulation as α and β are varied. In contrast, the transverse components retain the same intensity distribution over the Poincaré sphere, while the variation appears only in the polarization texture. Note that the factor $\sin^2 \theta_q$ in Eq. (2.38) vanishes near $\mathbf{q}_{\parallel} \approx 0$, producing a central null. As a result, the longitudinal component preserves a donut-shaped profile even when $\ell = \ell' = 0$ for both LG beams.

Motivated by this behavior, one may instead choose $\sigma' = -\sigma = -1$ and $\ell' = \ell = -1$, for which $\Delta J = (\ell' + \sigma') - (\ell + \sigma) = 0$. In this case, the longitudinal component becomes azimuthally uniform and can exhibit fully constructive or destructive interference over the entire angular spectrum when $(\alpha, \beta) = (0, \frac{\pi}{2})$ or $(\pi, \frac{\pi}{2})$, as shown in Figure 2.7.

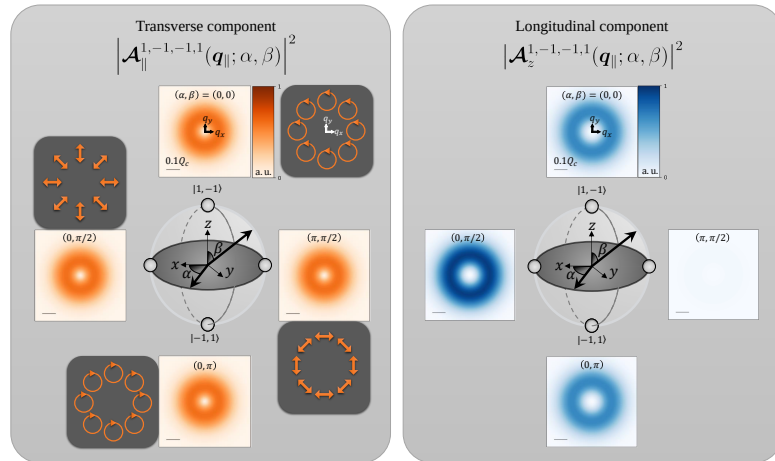


Figure 2.7 : Distribution (squared magnitude) of different components of $\mathcal{A}_{\parallel}^{1,-1,-1,-1}(\mathbf{q}_{\parallel}; \alpha, \beta)$ on the Poincaré sphere, together with the corresponding polarization states of the transverse components.

Chapter 3 Light-Matter Interaction

3.1 Theory of Light–Matter Interaction

3.1.1 Fermi’s golden rule

In this section, we examine the interaction between light and matter using **Fermi’s golden rule**. The transition rate for an optical excitation from the ground state $|GS\rangle$, with fully occupied valence bands, to an exciton state $|\Psi_{S,Q}^X\rangle$ with band index S and center-of-mass momentum Q is given by

$$\Gamma = \frac{2\pi}{\hbar} \left| \langle \Psi_{S,Q}^X | \hat{H}_I | GS \rangle \right|^2 \delta(E_{S,Q}^X - \hbar\omega). \quad (3.1)$$

The delta function enforces energy conservation between the exciton energy $E_{S,Q}^X$ and the photon energy $\hbar\omega$. Within the pseudospin model [6, 30, 41], the exciton state is written as

$$|\Psi_{S,Q}^X\rangle = \frac{1}{\sqrt{\Omega}} \sum_{v\mathbf{k}} \Lambda_{S,Q}(v\mathbf{k}) \hat{c}_{c,\mathbf{k}+Q}^\dagger \hat{h}_{v,-\mathbf{k}}^\dagger |GS\rangle. \quad (3.2)$$

Here, the particle operators $\hat{c}_{c,\mathbf{k}}^\dagger$ and $\hat{h}_{v,-\mathbf{k}}^\dagger$ create an electron in the conduction band with crystal momentum $\mathbf{k}$ and a hole in the valence band with crystal momentum $-\mathbf{k}$, respectively. The coefficient $\Lambda_{S,Q}(v\mathbf{k})$ represents the amplitude of the electron–hole configuration for exciton band S , and Ω is the area of the TMD monolayer. The matrix element in Eq. (3.1) can be expressed, within the **rotating wave approximation** [42], as

$$\langle \Psi_{S,Q}^X | \hat{H}_I | GS \rangle = \frac{1}{\sqrt{\Omega}} \sum_{v\mathbf{k}} \Lambda_{S,Q}^*(v\mathbf{k}) \langle \psi_{c,\mathbf{k}+Q} | H_I | \psi_{v,\mathbf{k}} \rangle, \quad (3.3)$$

where, for weak fields, the light–matter interaction Hamiltonian is given by [6, 30]

$$\hat{H}_I = \frac{|e|\hbar}{2m_e} \mathbf{A}(\mathbf{r}, t) \cdot \mathbf{p},$$

with $\mathbf{A}(\mathbf{r}, t)$ the vector potential in the **Coulomb gauge**, $\mathbf{p}$ the momentum operator, and m_e (e) the electron mass (charge). If the vector potential is expanded in a **plane-wave basis** as in Eq. (2.25), the matrix element in Eq. (3.3) becomes

$$\begin{aligned} \langle \psi_{c,\mathbf{k}+Q} | H_I | \psi_{v,\mathbf{k}} \rangle &= \frac{|e|\hbar}{2m_e} \sum_{\mathbf{q}_{\parallel}} \mathcal{A}(\mathbf{q}_{\parallel}) \left[\hat{\mathbf{e}} \cdot \langle \psi_{c,\mathbf{k}+Q} | e^{i\mathbf{q}_{\parallel} \cdot \mathbf{r}} \hat{\mathbf{p}} | \psi_{v,\mathbf{k}} \rangle \right] e^{i\omega t} \\ &\approx \frac{|e|\hbar}{2m_e} \sum_{\mathbf{q}_{\parallel}} \mathcal{A}(\mathbf{q}_{\parallel}) \left[\hat{\mathbf{e}} \cdot \delta_{\mathbf{q}_{\parallel}, Q} \langle \psi_{c,\mathbf{k}} | \hat{\mathbf{p}} | \psi_{v,\mathbf{k}} \rangle \right] e^{i\omega t}, \end{aligned} \quad (3.4)$$

where the approximation follows from the **electric dipole approximation**. In this approximation, the Bloch state wave functions are written as $\psi_{n,\mathbf{k}}(\mathbf{r}) = u_{n,\mathbf{k}}(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}$, and the plane-wave components combine to yield the Kronecker delta $\delta_{q\parallel,Q}$, enforcing the equality between the in-plane wavevector and the exciton center-of-mass momentum.

Substituting this result back, the matrix element in Eq. (3.3) [6, 30] becomes

$$\langle \Psi_{S,Q}^X | \hat{H}_I | GS \rangle = \frac{E_g}{2i\hbar} \mathcal{A}(\mathbf{Q}) (\hat{\epsilon} \cdot \mathbf{D}_{S,Q}^{X*}) e^{i\omega t}, \quad (3.5)$$

where $\mathbf{D}_{S,Q}^X$ is the exciton transition dipole moment, superscript * denotes complex conjugation, and E_g is the band gap at the K/K' valleys. This expression shows that optical excitation is governed by the alignment between the light polarization and the transition dipole moment. The transition rate is therefore proportional to both the magnitude of the transition dipole moment and the magnitude of the vector potential. Under the excitation of VVBs, the optical matrix element is defined as

$$\begin{aligned} \tilde{M}_{S,Q}^{\sigma,\ell,\sigma',\ell'} &\equiv \frac{1}{\sqrt{\Omega}} \sum_{v\mathbf{k}} \Lambda_{S,Q}^*(v\mathbf{k}) \left\langle \psi_{c,\mathbf{k}+\mathbf{Q}} \left| \frac{|e|}{2m_e} \mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha, \beta) \cdot \mathbf{p} \right| \psi_{v,\mathbf{k}} \right\rangle \\ &\approx \frac{E_g}{2i\hbar} \mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha, \beta) (\hat{\epsilon} \cdot \mathbf{D}_{S,Q}^{X*}), \end{aligned} \quad (3.6)$$

where the approximation follows from the electric dipole approximation. The corresponding Fermi's golden rule can then be written as [30]

$$\Gamma_{S,Q}^{\sigma,\ell,\sigma',\ell'} = \frac{2\pi}{\hbar} \left| \tilde{M}_{S,Q}^{\sigma,\ell,\sigma',\ell'} \right|^2 \rho(\hbar\omega = E_{S,Q}^X) \propto \left| \mathcal{A}^{\sigma,\ell,\sigma',\ell'}(\mathbf{Q}; \alpha, \beta) (\hat{\epsilon} \cdot \mathbf{D}_{S,Q}^X) \right|^2. \quad (3.7)$$

The transition dipole moment, obtained from density functional theory combined with the Bethe–Salpeter equation, is given in the small- $\mathbf{Q}$ limit by Ref [6, 30]:

Valley-mixed excitons (bright)

$$\begin{aligned} \text{Valley excitons (bright)} \quad & \mathbf{D}_{B\pm,Q}^X = \frac{D_B^X}{\sqrt{2}} (\cos \phi_Q \hat{x} \pm \sin \phi_Q \hat{y}), \\ \mathbf{D}_{K,Q}^X &= \frac{D_B^X}{\sqrt{2}} (\hat{x} + i\hat{y}), \\ \mathbf{D}_{K',Q}^X &= \frac{D_B^X}{\sqrt{2}} (\hat{x} - i\hat{y}) \end{aligned} \quad (3.8)$$

Gray excitons

$$\mathbf{D}_{G,Q}^X = D_G^X \hat{z}. \quad (3.10)$$

Here, the subscript symbols K , K' , $B\pm$, and G denote the K valley, K' valley, upper/ lower band, and gray exciton, respectively. The magnitude of the dipole moment of BXs (GX) is D_B^X (D_G^X) [6, 30]. From Eq. (3.7), it becomes clear why a single LG beam, whose TAM is encoded purely in its phase, does not produce a transition rate that reflects SOI or TAM. This is because the transition rate depends on the squared magnitude of the vector potential, which removes all phase information.

3.2 Manifestation of SOI in Excitonic Optical Transitions

3.2.1 Optical Transition Pattern of Excitons

We now can utilize the formulation above to analyze the SOI in excitonic optical transition for both valley-mixed BXs and GX under VVBs. The transition rates can be evaluated by substituting Eq. (2.33) and Eq. (2.34) into Eq. (3.7), yielding:

$$\begin{aligned}
 \left| \tilde{M}_{B+,Q}^{\sigma,\ell,\sigma',\ell'} \right|^2 &= \left(\frac{E_g D_B^X}{2\sqrt{2}\hbar} \right)^2 \left| \cos(\beta/2) \tilde{F}_{\ell,p}(Q) + e^{-i(\Delta J \phi_Q + \alpha)} \sin(\beta/2) \tilde{F}_{\ell',p}(Q) \right|^2, \\
 \left| \tilde{M}_{B-,Q}^{\sigma,\ell,\sigma',\ell'} \right|^2 &= \left(\frac{E_g D_B^X}{2\sqrt{2}\hbar} \right)^2 \left| \sigma \cos(\beta/2) \tilde{F}_{\ell,p}(Q) + e^{-i(\Delta J \phi_Q + \alpha)} \sigma' \sin(\beta/2) \tilde{F}_{\ell',p}(Q) \right|^2, \\
 \left| \tilde{M}_{G,Q}^{\sigma,\ell,\sigma',\ell'} \right|^2 &= \left(\frac{E_g D_G^X}{2\sqrt{2}\hbar} \right)^2 \sin^2 \theta_Q \left| \cos(\beta/2) \tilde{F}_{\ell,p}(Q) + e^{-i(\Delta J \phi_Q + \alpha)} \sin(\beta/2) \tilde{F}_{\ell',p}(Q) \right|^2,
 \end{aligned} \tag{3.11}$$

In this form, we observe that the SOI term $\Delta J = (\ell' + \sigma') - (\ell + \sigma)$ acts as an additional relative phase modulator with ϕ_Q dependence. Comparing the cases of valley-mixed BXs and GXs, Eq. (3.5) shows that the spin index σ can enter the phase of the optical matrix element. Consequently, the ϕ_Q dependence of the transition dipoles for valley-mixed BXs in Eq. (3.9) provides a mechanism for the emergence of ΔJ under excitation by a VVB state, whose polarization texture also depends on ϕ_Q . As a result, valley-mixed BXs can exhibit interference features analogous to those of GXs.

In contrast, for GXs, the SOI manifests directly through the interference term in Eq. (2.34). This originates from the fact that the transition dipole moment of GXs is oriented along the out-of-plane (z) direction and remains isotropic in the in-plane coordinates, making the interference the dominant mechanism responsible for revealing SOI with azimuthal structure. To explicitly reveal the interference effects, we expand the squared matrix elements for valley-mixed BXs as follows:

$$\begin{aligned}
 \left| \tilde{M}_{B+,Q}^{\sigma,\ell,\sigma',\ell'} \right|^2 &= \left(\frac{E_g D_B^X}{2\sqrt{2}\hbar} \right)^2 \left\{ \cos^2(\beta/2) \left| \tilde{F}_{\ell,p}(Q) \right|^2 + \sin^2(\beta/2) \left| \tilde{F}_{\ell',p}(Q) \right|^2 \right. \\
 &\quad \left. + \sin \beta \operatorname{Re} \left[\tilde{F}_{\ell,p}(Q) \tilde{F}_{\ell',p}^*(Q) e^{-i(\Delta J \phi_Q + \alpha)} \right] \right\}, \\
 \left| \tilde{M}_{B-,Q}^{\sigma,\ell,\sigma',\ell'} \right|^2 &= \left(\frac{E_g D_B^X}{2\sqrt{2}\hbar} \right)^2 \left\{ \cos^2(\beta/2) \left| \tilde{F}_{\ell,p}(Q) \right|^2 + \sin^2(\beta/2) \left| \tilde{F}_{\ell',p}(Q) \right|^2 \right. \\
 &\quad \left. + \sigma \sigma' \sin \beta \operatorname{Re} \left[\tilde{F}_{\ell,p}(Q) \tilde{F}_{\ell',p}^*(Q) e^{-i(\Delta J \phi_Q + \alpha)} \right] \right\}.
 \end{aligned} \tag{3.12}$$

Similarly, for GXs, we obtain:

$$\begin{aligned}
 \left| \tilde{M}_{G,Q}^{\sigma,\ell,\sigma',\ell'} \right|^2 &= \left(\frac{E_g D_G^X}{2\sqrt{2}\hbar} \right)^2 \sin^2 \theta_Q \left\{ \cos^2(\beta/2) \left| \tilde{F}_{\ell,p}(Q) \right|^2 + \sin^2(\beta/2) \left| \tilde{F}_{\ell',p}(Q) \right|^2 \right. \\
 &\quad \left. + \sin \beta \operatorname{Re} \left[\tilde{F}_{\ell,p}(Q) \tilde{F}_{\ell',p}^*(Q) e^{-i(\Delta J \phi_Q + \alpha)} \right] \right\}.
 \end{aligned} \tag{3.13}$$

In general, the transition rates for BXs and GXs exhibit interference structures analogous to the familiar interference term of the VVB state in Eq. (2.34). In addition, the transition rates of BXs take the same functional form when $\sigma = \sigma'$. Below, we discuss valley-mixed BXs and GX separately, since the physical origins of the SOI term are different. This distinction also affects how the manipulation of the transition rate should be interpreted. Note that one cannot simply set $\sigma = 0$ to represent linearly polarized light in this formulation, because the exponential form has been taken to simplify the derivation of the final bracket in Eq. (3.7).

Valley-mixed Bright Excitons : Polarization-Matching

For valley-mixed BXs, Eq. (3.12) shows that using opposite circular polarizations ($\sigma' = -\sigma$) leads to opposite interference behavior for the two valley-mixed branches. In addition, the SOI term ΔJ , which does not explicitly appear in the transverse intensity profile of the VVB itself, emerges once the phase factor generated by the dot product between the polarization unit vector and the transition dipole moment is included in Eq. (3.5).

Motivated by the polarization texture of the VVB state $|1, -1, -1, 1; \alpha, \beta\rangle$, shown in Figure 2.7, the condition $\Delta J = 0$ can be interpreted as a polarization-matching mechanism, illustrated in Figure 3.1. In particular, the polarization of the states

$$|1, -1, -1, 1; \alpha = 0, \beta = \pi/2\rangle \quad \text{and} \quad |1, -1, -1, 1; \alpha = \pi, \beta = \pi/2\rangle$$

matches the directions of the exciton transition dipoles D_{B+}^X and D_{B-}^X , respectively (see Figure 3.1). This enables selective excitation of the upper and lower exciton bands with the relative phase α as shown in Figure 3.2.

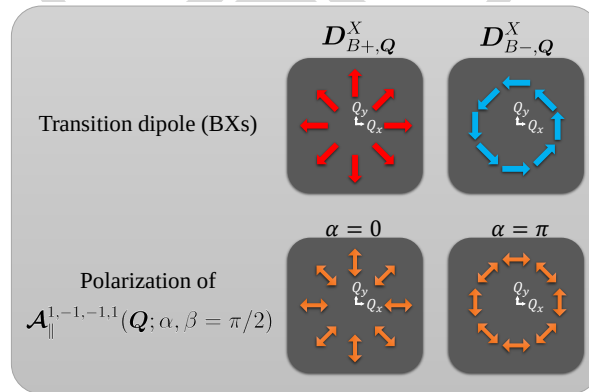


Figure 3.1 : Illustration of the transition dipole moments of valley-mixed BXs [6], and their matching with the transverse polarization of the VVB states $|1, -1, -1, 1; \alpha = 0, \beta = \pi/2\rangle$ and $|1, -1, -1, 1; \alpha = \pi, \beta = \pi/2\rangle$ in momentum space.

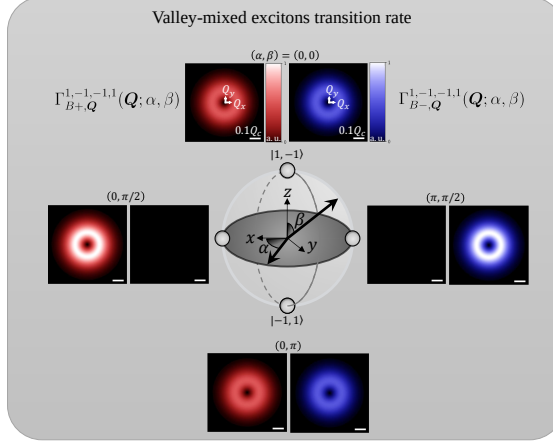


Figure 3.2 : Normalized transition rate of valley-mixed BXs excited by the VVB state $|1, -1, -1, 1; \alpha, \beta\rangle$ in momentum space.

However, since the valley splitting between valley-mixed BXs is only a few meV and is typically regarded as spectrally unresolvable [43], direct experimental verification of selective excitation may be difficult. Therefore, we consider the transition rate for the total BXs under the condition $(\sigma' = -\sigma)$, which is given by

$$\Gamma_{B\pm, \mathbf{Q}}^{\sigma, \ell, -\sigma, \ell'}(\alpha, \beta) \propto \sum_{S=B\pm} \left| \tilde{M}_{S, \mathbf{Q}}^{\sigma, \ell, -\sigma, \ell'} \right|^2 = \left(\frac{E_g D_B^X}{2\hbar} \right)^2 \left[\cos^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell, p}(Q) \right|^2 + \sin^2\left(\frac{\beta}{2}\right) \left| \tilde{F}_{\ell', p}(Q) \right|^2 \right]. \quad (3.14)$$

The formulation shows the cancellation of the two interference terms, leading to the disappearance of both the α dependence and the ΔJ dependence. The resulting distribution is shown in Figure 3.3, which exhibits an identical donut-shaped profile over the Poincaré sphere.

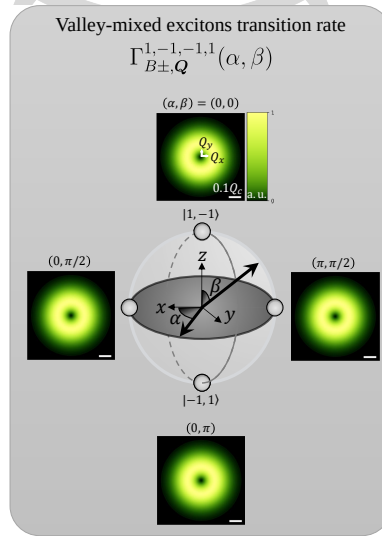


Figure 3.3 : Normalized transition rate of total BXs excited by the VVB state $|1, -1, -1, 1; \alpha, \beta\rangle$ in momentum space.

It should be noted that the condition $\Delta J = 0$ is not unique to this case. It can also be satisfied when $\ell' = \ell$ together with $\sigma' = \sigma$. However, in that situation the polarization-matching mechanism is absent. Instead, selective excitation is achieved simply by tuning the relative superposition parameters, effectively switching the excitation channel on and off, as shown in Figure 3.4 and Figure 3.5. Consequently, the excitation behavior of the two BX branches becomes identical, consistent with Eq. (3.12).

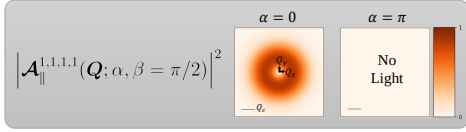


Figure 3.4 : Distribution of the transverse component for the VVB state $|1, 1, 1, 1; \alpha, \beta\rangle$ in momentum space.

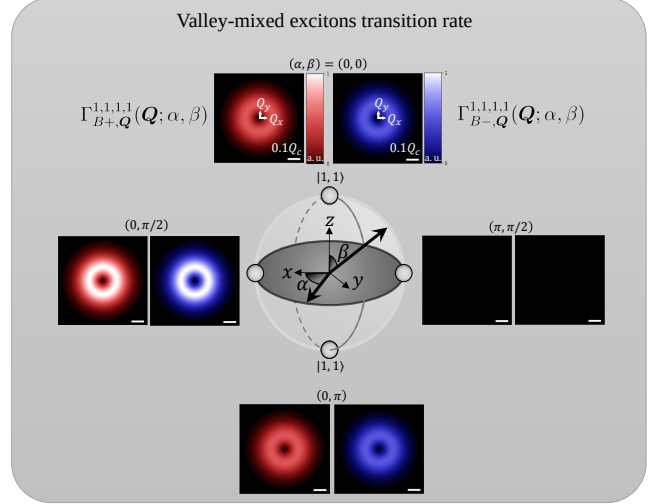


Figure 3.5 : Normalized transition rate of valley-mixed BXs excited by the VVB state $|1, 1, 1, 1; \alpha, \beta\rangle$ in momentum space.

Gray Excitons: Manifestation of SOI

For gray excitons, whose transition dipole moments possess only a z component, no additional phase factor is generated by the dot product between the polarization vector and the transition dipole moment in Eq. (3.5). Therefore, the SOI term ΔJ in Eq. (3.13) originates entirely from the longitudinal component of the VVB state in Eq. (2.34). As a result, GX provide an ideal probe of the SOI of light, since their transition rates directly reflect the intrinsic angular-momentum structure of the optical field. At the same time, they offer a possible route for exciting GX using paraxial structured beams.

Further exploit the intensity texture of the VVB state $|1, -1, -1, 1; \alpha, \beta\rangle$, shown in Figure 2.7. In particular, complete switching of the excitation on and off can be realized by tuning the relative phase between $\alpha = 0$ and $\alpha = \pi$ under excitation by the state

$$|1, -1, -1, 1; \alpha, \beta = \pi/2\rangle,$$

as shown in Figure 3.6.

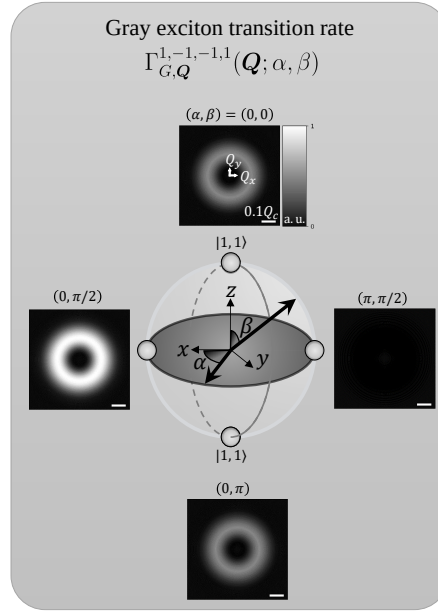


Figure 3.6 : Normalized transition rate of GXs excited by the VVB state $|1, -1, -1, 1; \alpha, \beta\rangle$ in momentum space.

Beyond complete switching, one may also use the case $\Delta J = 1$ to excite gray excitons with selected finite center-of-mass momentum. For example, consider the state

$$|1, -1, -1, 2; \alpha, \beta = \pi/2\rangle,$$

for which

$$\Delta J = (\ell' + \sigma') - (\ell + \sigma) = 1.$$

First, the radial factor $\tilde{F}_{\ell p}(q_{\parallel})$ of the two constituent LG beams are no longer identical, so the transition rates at $\beta = 0$ and $\beta = \pi$ (the north and south poles of the Poincaré sphere) is different. Figure 3.7 shows that the LG beam with larger ℓ produces a larger-radius donut-shaped transition-rate profile in the momentum space.

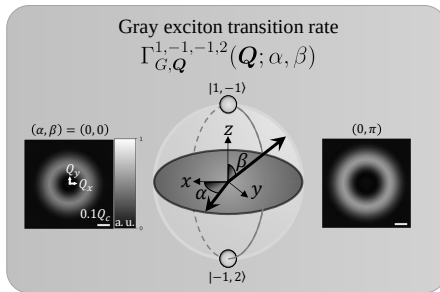


Figure 3.7 : Normalized transition rate of GXs excited by the VVB state $|1, -1, -1, 2; \alpha, \beta\rangle$ in momentum space at the north/south poles of the Poincaré sphere.

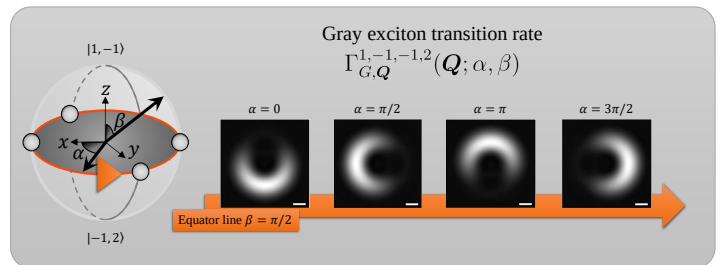


Figure 3.8 : Normalized transition rate of GXs excited by the VVB state $|1, -1, -1, 2; \alpha, \beta\rangle$ in momentum space along the equator of the Poincaré sphere. The scale and color bars are the same as in Figure 3.7.

Second, although the two LG beams do not fully overlap, the condition $\Delta J = 1$ still induces one full interference cycle in the azimuthal direction. Consequently, by tuning α , one can selectively excite GX within a finite range of center-of-mass momentum at $\beta = \pi/2$ (the equator of the Poincaré sphere), as shown in Figure 3.8.

3.2.2 Selective Excitation of Excitons

By summing the contributions in momentum space excited by the VVBs, we obtain the total transition rate, which represents the intensity of the material absorption. Under the paraxial approximation, where $\theta_Q = \frac{Q}{\sqrt{Q^2 + q_0^2}} \approx \frac{Q}{q_0}$ is taken to simplify the integral, the total transition rate can be evaluated as:

$$\Gamma_S^{\sigma, \ell, \sigma', \ell'} \equiv \sum_Q \Gamma_{S, Q}^{\sigma, \ell, \sigma', \ell'} = \frac{\Omega}{(2\pi)^2} \int_0^\infty \int_0^{2\pi} \Gamma_{S, Q}^{\sigma, \ell, \sigma', \ell'} Q d\phi_Q dQ.$$

The total transition rates for the upper-band BXs, lower-band BXs, and GXs are obtained as

$$\begin{aligned} \Gamma_{B+}^{\sigma, \ell, \sigma', \ell'}(\alpha, \beta) &\propto \left| \tilde{M}_{B+}^{\sigma, \ell, \sigma', \ell'} \right|^2 \\ &= \frac{\Phi}{8} \left(\frac{E_g D_B^X}{a_0 \Omega \hbar} \right)^2 \left\{ 1 + \delta_{\Delta J, 0} \sin \beta \cos \left[\frac{\pi}{2} (|\ell'| - |\ell|) + \alpha \right] \frac{\left(\frac{|\ell'| + |\ell|}{2} \right)!}{\sqrt{|\ell'|! |\ell|!}} \right\}, \\ \Gamma_{B-}^{\sigma, \ell, \sigma', \ell'}(\alpha, \beta) &\propto \left| \tilde{M}_{B-}^{\sigma, \ell, \sigma', \ell'} \right|^2 \\ &= \frac{\Phi}{8} \left(\frac{E_g D_B^X}{a_0 \Omega \hbar} \right)^2 \left\{ 1 + \delta_{\Delta J, 0} \sigma \sigma' \sin \beta \cos \left[\frac{\pi}{2} (|\ell'| - |\ell|) + \alpha \right] \frac{\left(\frac{|\ell'| + |\ell|}{2} \right)!}{\sqrt{|\ell'|! |\ell|!}} \right\}, \\ \Gamma_G^{\sigma, \ell, \sigma', \ell'}(\alpha, \beta) &\propto \left| \tilde{M}_G^{\sigma, \ell, \sigma', \ell'} \right|^2 \\ &= \frac{\Phi}{4} \left(\frac{E_g D_G^X}{a_0 q_0 W_0 \Omega \hbar} \right)^2 \left\{ \left(1 + \cos^2 \frac{\beta}{2} |\ell| + \sin^2 \frac{\beta}{2} |\ell'| \right) \right. \\ &\quad \left. + \delta_{\Delta J, 0} \sin \beta \cos \left[\frac{\pi}{2} (|\ell'| - |\ell|) + \alpha \right] \frac{\left(\frac{|\ell'| + |\ell|}{2} + 1 \right)!}{\sqrt{|\ell'|! |\ell|!}} \right\}, \end{aligned} \quad (3.15)$$

where the constant radiant power is $\Phi = (a_0 A_0^{\text{LG}} W_0)^2$, and $a_0 = \frac{\omega}{\sqrt{2c\mu_0}}$.

At first glance, the SOI appears to be absent, since ℓ and σ are not explicitly coupled in the second term of all three expressions. However, the trigonometric factor can be reinterpreted as

$$\sin \beta \cos \left[\frac{\pi}{2} (|\ell'| - |\ell|) + \alpha \right] = (\text{Re} [e^{i(|\ell'| - |\ell|)}], \text{Im} [e^{i(|\ell'| - |\ell|)}], 0) \cdot \begin{pmatrix} \sin \beta \cos \alpha \\ \sin \beta \sin \alpha \\ \cos \beta \end{pmatrix}, \quad (3.16)$$

which reveals its origin as a phase correction between the complex components of two VVB states located on the poles of Poincaré sphere.

More specifically, the complex phase of the radial factor $\tilde{F}_{\ell,p}(Q)$ takes the form $(-i)^{|\ell|}$. As a result, the interference is governed by the difference in $|\ell|$, which introduces a phase shift across whole momentum space, together with the VVB state specified by (α, β) .

At the same time, the contribution from SOI (or, for BXs, the polarization-matching mechanism) is encoded through $\delta_{\Delta J,0}$. When $\Delta J \neq 0$, the resulting azimuthal phase mismatch—such as that illustrated in Figure 3.8—leads to destructive averaging over momentum space, effectively given no net contribution to the total transition rate.

From this perspective, the factorial term in Eq. (3.15) can be interpreted as a geometric correction arising from the mismatch between the radial factors $\tilde{F}_{\ell,p}(Q)$ of the two LG beams. In particular, when $|\ell| = |\ell'|$, it reduces to simple constants, yielding 1 for BXs and $1 + |\ell|$ for GXs. These values are consistent with the corresponding prefactors in the non-interference contribution (i.e., the term independent of $\delta_{\Delta J,0}$).

As for the beam waist W_0 , which appears in the denominator of the prefactor for the GX total transition rate, it reflects that smaller W_0 enhances the longitudinal field component and thus increases the excitation efficiency of GXs.

On the other hand, the total transition rate for the total BXs under the condition $(\sigma' = -\sigma)$ can be similarly written as

$$\Gamma_{B\pm}^{\sigma,\ell,-\sigma,\ell'}(\alpha, \beta) \propto \sum_{S=B\pm} \left| \tilde{M}_S^{\sigma,\ell,-\sigma,\ell'} \right|^2 = \frac{\Phi}{4} \left(\frac{E_g D_B^X}{a_0 \Omega \hbar} \right)^2. \quad (3.17)$$

Therefore, one can exploit both Eq. (3.15) and Eq. (3.17) to analyze the optical transition rates. In particular, we focus on the equator of the Poincaré sphere ($\beta = \pi/2$), where the VVB corresponds to a maximal superposition of two LG beams. By varying the azimuthal angle α along the equator, we examine the dependence of the total transition rate on ℓ (with $|\ell| = |\ell'|$) using the VVB states $|1, \ell, -1, -\ell; \alpha, \beta = \pi/2\rangle$, as shown in Figure 3.9.

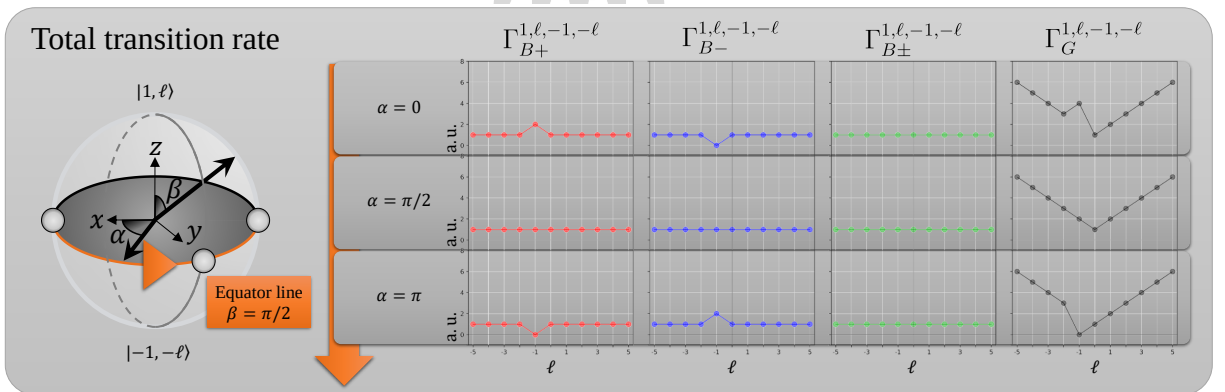


Figure 3.9 : Normalized total transition rates of BXs, total BXs, and GX excited by a VVB state $|1, \ell, -1, -\ell; \alpha, \beta = \pi/2\rangle$, for $\ell \in \mathbb{Z}$ with $\ell \in [-5, 5]$, and for azimuthal angles $\alpha = 0, \pi/2, \pi$.

In Figure 3.9, the total transition rate for each exciton exhibits a clear dependence on α when the condition $\Delta J = 0$ is satisfied at $\ell = -1$. As shown within the same column, varying

α from 0 to $\pi/2$ to π leads to modulation of the transition rate. This behavior is consistent with the results in Figure 3.2 and Figure 3.6, where the transition rate exhibits fully constructive or destructive interference over momentum space at $\alpha = 0$ or $\alpha = \pi$.

In contrast, the total transition rate of total BXs becomes independent of α and ℓ . This is due to the cancellation between the contributions of the upper and lower BX bands, as reflected in Figure 3.5.

Apart from this special cancellation, BXs generally exhibit independence from ℓ , unlike GXs. This difference originates from the additional Q -dependent prefactor in the GX case, specifically $\sin \theta_Q \approx Q/q_0$ in Eq. (3.13). Consequently, Eq. (3.15) shows a dependence on ℓ even in the absence of interference effects.

Motivated by the special properties of the VVB state at the equator of the Poincaré sphere, $|1, -1, -1, 1; \alpha, \beta = \pi/2\rangle$, one can achieve smooth and selective excitation over half of the equator ($\alpha = 0$ to π), as shown in Figure 3.10.

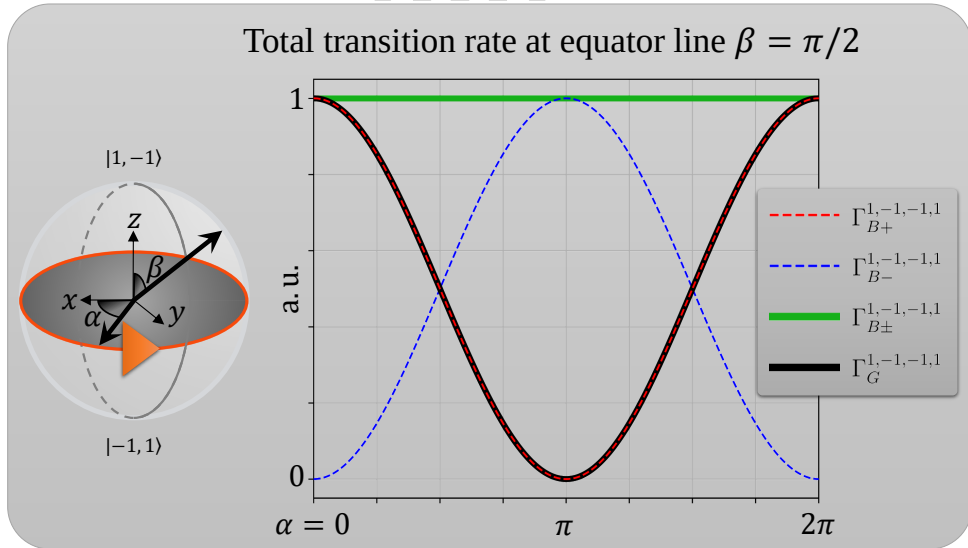


Figure 3.10 : Normalized total transition rates of BXs, total BXs, and GXs excited by a VVB state $|1, -1, -1, 1; \alpha, \beta = \pi/2\rangle$ for azimuthal angles $\alpha = 0$ to 2π .

These observations originate from SOI associated with the longitudinal component of the LG beam, together with a polarization-matching mechanism. The former arises from the gauge transformation, specifically the second term in Eq. (2.23), as well as the intrinsic structure of the LG beam. The latter originates from the correspondence between the transition dipole structure of the upper and lower BX bands and the polarization textures of the VVB state. In both cases, Fermi's golden rule provides a theoretical framework for interpreting these results and establishes a basis for further studies involving different VVB configurations, excitonic species, and other comprehensive investigations.

Chapter 4 Exciton Wave Packet States

4.1 Superposition of Finite-Momentum Exciton (SFME) states

4.1.1 Global Phase in Valley-Mixed Exciton

In this section, we establish the theoretical foundation for further studies of exciton wave packets. A wave packet of the particle refers to a superposition of momentum eigenstates, where the carrier wave (ripples) possesses a characteristic wavelength determined by the dominant momentum component, while the envelope describes the finite spatial extent of the particle. The velocity associated with the envelope, known as the group velocity, corresponds to the particle velocity in classical physics. From this perspective, the envelope of a wave packet provides a good intuition of the particle's position and motion [42, 44].

Using time-dependent perturbation theory [42], the time-dependent exciton state under weak photoexcitation can be expressed as a superposition of finite-momentum exciton (SFME) states [6]

$$|\Psi^X(t)\rangle = |GS\rangle + \sum_{S=B\pm} \sum_{\mathbf{Q}} \tilde{c}_{S,\mathbf{Q}}^{(1)}(t) e^{-i\omega_{S,\mathbf{Q}}t} |\Psi_{S,\mathbf{Q}}^X\rangle, \quad (4.1)$$

which consists of a linear combination of the ground state $|GS\rangle$ and exciton states $|\Psi_{S,\mathbf{Q}}^X\rangle$. Within first-order perturbation theory, the amplitude coefficients take the form

$$\tilde{c}_{S,\mathbf{Q}}^{(1)}(t) = \tilde{\gamma}_{S,\mathbf{Q}}(t) \left(\frac{\tilde{M}_{S,\mathbf{Q}}}{\hbar\omega_{S,\mathbf{Q}}} \right),$$

where the time dependence coefficient $\tilde{\gamma}_{S,\mathbf{Q}}(t) \approx \tilde{\gamma}_0(t)$ and the natural frequency $\omega_{S,\mathbf{Q}} = E_{S,\mathbf{Q}}^X/\hbar \approx E_{S,0}^X/\hbar = \omega_0$. The amplitude coefficients can be separated into time-dependent and time-independent parts, where the latter contains the optical matrix element introduced in Eq. (3.6). For simplicity, we consider only excitation by a single LG beam (with radial mode index $p = 0$), such that the optical matrix element takes the form

$$\tilde{M}_{S,\mathbf{Q}}^{\hat{\epsilon},\ell} \approx \frac{E_g}{2i\hbar} \mathcal{A}^{\hat{\epsilon},\ell}(\mathbf{Q}) (\hat{\epsilon} \cdot \mathbf{D}_{S,\mathbf{Q}}^{X*}). \quad (4.2)$$

Here, derivations analogous to those in the previous chapter—including the rotating-wave approximation and electric dipole approximation—are omitted for brevity. By adopting the polarization vector $\hat{\epsilon}$ instead of photon helicity σ , we relax the restriction to circular polarization. The vector potential is denoted by $\mathcal{A}^{\hat{\epsilon},\ell}(\mathbf{q}_{\parallel}) = \mathcal{A}_{\ell p}^C(\mathbf{q}_{\parallel})$, which corresponds to the complex amplitude introduced in Eq. (2.29).

In the work of Guan-Hao Peng *et al.* [6], the analysis primarily focuses on the static part

of the envelope function, which is obtained by constructing the envelope function of the SFME state using only the amplitude coefficients,

$$F_S^{\hat{\epsilon},\ell}(\mathbf{R}_c, t) = \frac{1}{\sqrt{\Omega}} \sum_Q \tilde{c}_{S,Q}^{(1)}(t) e^{i\mathbf{Q} \cdot \mathbf{R}_c}. \quad (4.3)$$

To explicitly isolate the static contribution, we use the relation $F_S^{\hat{\epsilon},\ell}(\mathbf{R}_c, t) \approx \tilde{\gamma}_0(t) f_S^{\hat{\epsilon},\ell}(\mathbf{R}_c)$, which defines the static part of the envelope function

$$f_S^{\hat{\epsilon},\ell}(\mathbf{R}_c) = \frac{1}{\hbar\omega_0\sqrt{\Omega}} \sum_Q \tilde{M}_{S,Q}^{\hat{\epsilon},\ell} e^{i\mathbf{Q} \cdot \mathbf{R}_c}. \quad (4.4)$$

However, the valley-mixed exciton state obtained under pseudo-spin model [6],

$$|\Psi_{\pm,Q}^X\rangle = e^{i\delta(Q)} (e^{-i\phi_Q} |\Psi_{K,Q}^X\rangle \pm e^{i\phi_Q} |\Psi_{K',Q}^X\rangle) / \sqrt{2}, \quad (4.5)$$

naturally carries a Q -dependent global phase $\delta(Q)$ allowed by the inner structure. Consequently, the optical matrix element acquires the same phase factor,

$$\tilde{M}_{\pm,Q}^{\hat{\epsilon},\ell} = e^{i\delta(Q)} (e^{-i\phi_Q} \tilde{M}_{K,Q}^{\hat{\epsilon},\ell} \pm e^{i\phi_Q} \tilde{M}_{K',Q}^{\hat{\epsilon},\ell}) / \sqrt{2}. \quad (4.6)$$

As a result, the global phase eventually modify the resulting envelope function across momentum space when the inner structure of exciton wave function is not taken into consideration.

4.1.2 Zeroth-Order Approximation

To properly address this issue without investing a large amount of computational resources in constructing the exciton wave function, one can instead analytically extract its Q -dependent phase structure. We begin by considering the internal structure of the exciton wave function [45, 46],

$$\Psi_{S,Q}^X(\mathbf{r}_e, \mathbf{r}_h) = \sum_{v\mathbf{k}} \Lambda_{S,Q}(v\mathbf{k}) \psi_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{r}_e) \psi_{v,\mathbf{k}}^*(\mathbf{r}_h), \quad (4.7)$$

where $\psi_{c,\mathbf{k}}(\mathbf{r}_e)$ and $\psi_{v,\mathbf{k}}(\mathbf{r}_h)$ denote the single-particle Bloch wave functions for an electron in the conduction band and a hole in the valence band, respectively. Here, $\mathbf{r}_e$ and $\mathbf{r}_h$ represent the electron and hole position vectors. Other notations follow Eq. (3.2).

For the purpose of this extraction, it is convenient to rewrite the exciton wave function in terms of a weighted-average position vector $\mathbf{R}_c$ ¹ and the relative position vector $\mathbf{r}$:

$$\begin{aligned} \mathbf{R}_c &= a\mathbf{r}_e + b\mathbf{r}_h, & \mathbf{r} &= \mathbf{r}_e - \mathbf{r}_h, & a + b &= 1, \\ \mathbf{r}_e &= \mathbf{R}_c + b\mathbf{r}, & \mathbf{r}_h &= \mathbf{R}_c - a\mathbf{r}. \end{aligned} \quad (4.8)$$

¹ Although the physical meaning of this weighted-average position vector is not strictly identical to the conventional center-of-mass position vector in classical physics, we retain the subscript “c” to avoid confusion with the Bravais lattice vector $\mathbf{R}$.

Using these relations, the exciton wave function can be rewritten as

$$\Psi_{S,Q}^X(\mathbf{R}_c, \mathbf{r}) = \sum_{v\mathbf{c}\mathbf{k}} \Lambda_{S,Q}(v\mathbf{c}\mathbf{k}) \psi_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{R}_c + b\mathbf{r}) \psi_{v,\mathbf{k}}^*(\mathbf{R}_c - a\mathbf{r}), \quad (4.9)$$

In addition, Bloch's theorem states that, in a material with a periodic lattice structure—such as a two-dimensional TMD characterized by Bravais lattice vectors $\mathbf{R}$ —the single-particle wave function satisfies [42, 46]

$$\psi_{n,\mathbf{k}}(\mathbf{r}) = u_{n,\mathbf{k}}(\mathbf{r}) e^{i\mathbf{k}\cdot\mathbf{r}}, \quad (4.10)$$

where the wave function is expressed as the product of a cell-periodic function, $u_{n,\mathbf{k}}(\mathbf{r}) = u_{n,\mathbf{k}}(\mathbf{r} + \mathbf{R})$, and a plane-wave component $e^{i\mathbf{k}\cdot\mathbf{r}}$. Using this property, the exciton wave function can be rewritten as

$$\begin{aligned} \Psi_{S,Q}^X(\mathbf{R}_c, \mathbf{r}) &= \sum_{v\mathbf{c}\mathbf{k}} \Lambda_{S,Q}(v\mathbf{c}\mathbf{k}) \left[u_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{R}_c + b\mathbf{r}) e^{i(\mathbf{k}+\mathbf{Q})\cdot(\mathbf{R}_c+b\mathbf{r})} u_{v,\mathbf{k}}^*(\mathbf{R}_c - a\mathbf{r}) e^{-i\mathbf{k}\cdot(\mathbf{R}_c-a\mathbf{r})} \right] \\ &\Rightarrow \sum_{v\mathbf{c}\mathbf{k}} \Lambda_{S,Q}(v\mathbf{c}\mathbf{k} - b\mathbf{Q}) e^{i\mathbf{k}\cdot\mathbf{r}} e^{i\mathbf{Q}\cdot\mathbf{R}_c} \left[u_{c,\mathbf{k}+a\mathbf{Q}}(\mathbf{R}_c + b\mathbf{r}) u_{v,\mathbf{k}-b\mathbf{Q}}^*(\mathbf{R}_c - a\mathbf{r}) \right]. \end{aligned} \quad (4.11)$$

The second arrow is obtained by shifting crystal momentum by $-b\mathbf{Q}$, which allows the $\mathbf{Q}$ -dependent phase originating from the plane-wave components to be written in the compact form $e^{i\mathbf{Q}\cdot\mathbf{R}_c}$.

Finally, we adopt the convention

$$a = 1, \quad b = 0,$$

which gives the exciton wave function in the form

$$\Psi_{S,Q}^X(\mathbf{R}_c, \mathbf{r}) = \sum_{v\mathbf{c}\mathbf{k}} \Lambda_{S,Q}(v\mathbf{c}\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}} e^{i\mathbf{Q}\cdot\mathbf{R}_c} \left[u_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{R}_c) u_{v,\mathbf{k}}^*(\mathbf{R}_c - \mathbf{r}) \right]. \quad (4.12)$$

Under this convention, the hole cell-periodic function becomes independent of $\mathbf{Q}$.

Further simplification can be achieved by performing a **Taylor expansion** around $\mathbf{Q} = 0$ for the implicitly $\mathbf{Q}$ -dependent terms, namely $\Lambda_{S,Q}(v\mathbf{c}\mathbf{k})$ and $u_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{R}_c)$. Explicitly,

$$\begin{aligned} \Lambda_{S,Q}(v\mathbf{c}\mathbf{k}) &= \sum_{m=0}^{\infty} \frac{1}{m!} \frac{d^m}{d\mathbf{k}^m} \Lambda_{S,Q}(v\mathbf{c}\mathbf{k}) \cdot \mathbf{Q}^m = \Lambda_{S,0}(v\mathbf{c}\mathbf{k}) + \mathbf{Q} \cdot \nabla_{\mathbf{k}} \Lambda_{S,0}(v\mathbf{c}\mathbf{k}) + \dots \\ u_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{R}_c) &= \sum_{m=0}^{\infty} \frac{1}{m!} \frac{d^m}{d\mathbf{k}^m} u_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{R}_c) \cdot \mathbf{Q}^m = u_{c,\mathbf{k}}(\mathbf{R}_c) + \mathbf{Q} \cdot \nabla_{\mathbf{k}} u_{c,\mathbf{k}}(\mathbf{R}_c) + \dots \end{aligned} \quad (4.13)$$

Their product therefore becomes

$$\begin{aligned}
\Lambda_{S,Q}(v\mathbf{c}\mathbf{k})u_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{R}_c) &= \underbrace{\Lambda_{S,0}(v\mathbf{c}\mathbf{k})u_{c,\mathbf{k}}(\mathbf{R}_c)}_{\text{zeroth order}} \\
&+ \underbrace{\Lambda_{S,0}(v\mathbf{c}\mathbf{k})[\mathbf{Q} \cdot \nabla_{\mathbf{k}}u_{c,\mathbf{k}}(\mathbf{R}_c)] + [\mathbf{Q} \cdot \nabla_{\mathbf{k}}\Lambda_{S,0}(v\mathbf{c}\mathbf{k})]u_{c,\mathbf{k}}(\mathbf{R}_c)}_{\text{first order}} \\
&+ \underbrace{[\mathbf{Q} \cdot \nabla_{\mathbf{k}}\Lambda_{S,0}(v\mathbf{c}\mathbf{k})][\mathbf{Q} \cdot \nabla_{\mathbf{k}}u_{c,\mathbf{k}}(\mathbf{R}_c)]}_{\text{second order}} \\
&+ \text{higher-order terms.}
\end{aligned} \tag{4.14}$$

Substituting this expansion into Eq. (4.12), we denote the contribution from the m th-order term by $\Psi_{S,Q}^{X(m)}(\mathbf{R}_c, \mathbf{r})$:

$$\begin{aligned}
\Psi_{S,Q}^X(\mathbf{R}_c, \mathbf{r}) &= e^{i\mathbf{Q} \cdot \mathbf{R}_c} \sum_{v\mathbf{c}\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} \left\{ \Lambda_{S,Q}(v\mathbf{c}\mathbf{k})u_{c,\mathbf{k}+\mathbf{Q}}(\mathbf{R}_c) \right\} u_{v,\mathbf{k}}^*(\mathbf{R}_c - \mathbf{r}) \\
&= e^{i\mathbf{Q} \cdot \mathbf{R}_c} \sum_{v\mathbf{c}\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} \left[\Psi_{S,Q}^{X(0)}(\mathbf{R}_c, \mathbf{r}) + \Psi_{S,Q}^{X(1)}(\mathbf{R}_c, \mathbf{r}) + \Psi_{S,Q}^{X(2)}(\mathbf{R}_c, \mathbf{r}) + \cdots \right] \\
&\approx e^{i\mathbf{Q} \cdot \mathbf{R}_c} \sum_{v\mathbf{c}\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} \Lambda_{S,0}(v\mathbf{c}\mathbf{k})u_{c,\mathbf{k}}(\mathbf{R}_c)u_{v,\mathbf{k}}^*(\mathbf{R}_c - \mathbf{r}),
\end{aligned} \tag{4.15}$$

where the final approximation corresponds to the zeroth-order² approximation in the small- $\mathbf{Q}$ limit and extract the $\mathbf{Q}$ -dependent phase term [47] by having

$$\Psi_{S,Q}^X(\mathbf{R}_c, \mathbf{r}) \approx e^{i\mathbf{Q} \cdot \mathbf{R}_c} \Psi_{S,0}^X(\mathbf{R}_c, \mathbf{r}) \tag{4.16}$$

4.1.3 Exciton Wave Packet State

With the relation in Eq. (4.16), the valley exciton wave functions

$$\Psi_{K/K',Q}^X(\mathbf{R}_c, \mathbf{r}) \approx e^{i\mathbf{Q} \cdot \mathbf{R}_c} \Psi_{K/K',0}^X(\mathbf{R}_c, \mathbf{r}) \tag{4.17}$$

can form the valley-mixed exciton wave functions under the pseudo-spin model [6]

$$\begin{aligned}
\Psi_{\pm,Q}^X(\mathbf{R}_c, \mathbf{r}) &\approx \frac{1}{\sqrt{2}} [e^{-i\phi_Q} \Psi_{K,Q}^X(\mathbf{R}_c, \mathbf{r}) \pm e^{i\phi_Q} \Psi_{K',Q}^X(\mathbf{R}_c, \mathbf{r})] \\
&\approx \frac{e^{i\mathbf{Q} \cdot \mathbf{R}_c}}{\sqrt{2}} [e^{-i\phi_Q} \Psi_{K,0}^X(\mathbf{R}_c, \mathbf{r}) \pm e^{i\phi_Q} \Psi_{K',0}^X(\mathbf{R}_c, \mathbf{r})]
\end{aligned} \tag{4.18}$$

Considering the internal structure, the envelope function is constructed as a weighted superposition of exciton states, represented in real space by their corresponding wave functions,

²A similar expansion method is used in evaluating the electron-hole exchange interaction (EHEI) within the Bethe-Salpeter equation, where the first-order approximation is typically retained. At first glance, this may appear inconsistent with the zeroth-order approximation adopted here. However, in the case of the EHEI, the zeroth-order term gives no contribution (roughly speaking), making a higher-order approximation necessary.

with amplitude coefficients derived from first-order time-dependent perturbation theory

$$\begin{aligned}
F_S^{\hat{\epsilon},\ell}(\mathbf{R}_c, \mathbf{r}, t) &= \sum_{\mathbf{Q}} \tilde{c}_{S,\mathbf{Q}}^{(1)}(t) e^{-i\omega_{S,\mathbf{Q}}t} \Psi_{S,\mathbf{Q}}^X(\mathbf{R}_c, \mathbf{r}) \\
&\approx \sum_{\mathbf{Q}} \tilde{c}_{S,\mathbf{Q}}^{(1)}(t) e^{-i\omega_{S,\mathbf{Q}}t} e^{i\mathbf{Q} \cdot \mathbf{R}_c} \Psi_{S,0}^X(\mathbf{R}_c, \mathbf{r}).
\end{aligned} \tag{4.19}$$

Similarly, using the relation $F_S^{\hat{\epsilon},\ell}(\mathbf{R}_c, \mathbf{r}, t) \approx \tilde{\gamma}_0(t) e^{-i\omega_0 t} f_S^{\hat{\epsilon},\ell}(\mathbf{R}_c, \mathbf{r})$, we define the static part of envelope function as

$$f_S^{\hat{\epsilon},\ell}(\mathbf{R}_c, \mathbf{r}) = (\hbar\omega_0)^{-1} \Psi_{S,0}^X(\mathbf{R}_c, \mathbf{r}) \sum_{\mathbf{Q}} \tilde{M}_{S,\mathbf{Q}}^{\hat{\epsilon},\ell} e^{i\mathbf{Q} \cdot \mathbf{R}_c}. \tag{4.20}$$



Chapter 5 Summary and outlook



References

- [1] Nicolas Roisin et al. “Phonon-limited mobility for electrons and holes in highly-strained silicon”. In: *npj Computational Materials* 10.1 (2024), p. 242. DOI: 10.1038/s41524-024-01425-0.
- [2] Aaron Bostwick et al. “Quasiparticle dynamics in graphene”. In: *Nature Physics* 3.1 (2007), pp. 36–40. DOI: 10.1038/nphys477.
- [3] Andrea Splendiani et al. “Emerging Photoluminescence in Monolayer MoS₂”. In: *Nano Letters* 10.4 (2010), pp. 1271–1275. DOI: 10.1021/nl903868w.
- [4] Alexander Krasnok and Andrea Alù. “Valley-Selective Response of Nanostructures Coupled to 2D Transition-Metal Dichalcogenides”. In: *Applied Sciences* 8.7 (2018), p. 1157. DOI: 10.3390/app8071157.
- [5] Guan-Hao Peng et al. “Distinctive Signatures of the Spin- and Momentum-Forbidden Dark Exciton States in the Photoluminescence of Strained WSe₂ Monolayers under Thermalization”. In: *Nano Letters* 19.4 (2019), pp. 2299–2312. DOI: 10.1021/acs.nanolett.8b04786.
- [6] Guan-Hao Peng et al. “Tailoring the superposition of finite-momentum valley exciton states in transition-metal dichalcogenide monolayers by using polarized twisted light”. In: *Physical Review B* 106.15 (2022), p. 155304. DOI: 10.1103/PhysRevB.106.155304.
- [7] Lorenz Maximilian Schneider et al. “Direct Measurement of the Radiative Pattern of Bright and Dark Excitons and Exciton Complexes in Encapsulated Tungsten Diselenide”. In: *Scientific Reports* 10.1 (2020), p. 8091. DOI: 10.1038/s41598-020-64838-z.
- [8] J. H. Poynting. “The Wave Motion of a Revolving Shaft, and a Suggestion as to the Angular Momentum in a Beam of Circularly Polarised Light”. In: *Proceedings of the Royal Society of London A* 82.557 (1909), pp. 560–567.
- [9] Richard A. Beth. “Mechanical Detection and Measurement of the Angular Momentum of Light”. In: *Physical Review* 50.2 (1936), pp. 115–125. DOI: 10.1103/PhysRev.50.115.
- [10] L. Allen et al. “Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes”. In: *Physical Review A* 45.11 (1992), pp. 8185–8189. DOI: 10.1103/PhysRevA.45.8185.
- [11] J. M. Jauch and F. Rohrlich. *The theory of photons and electrons: The relativistic quantum field theory of charged particles with spin one-half*. 2nd ed. Berlin: Springer-Verlag, 1976. DOI: 10.1007/978-3-642-80951-4.
- [12] Andrew Forbes, Light Mkhumbuzza, and Liang Feng. “Orbital angular momentum lasers”. In: *Nature Reviews Physics* 6.6 (2024), pp. 352–364. DOI: 10.1038/s42254-024-00715-2.
- [13] Manuel Erhard et al. “Twisted photons: new quantum perspectives in high dimensions”. In: *Light: Science & Applications* 7.3 (2018), p. 17146. DOI: 10.1038/lsa.2017.146.

- [14] H. He et al. “Direct Observation of Transfer of Angular Momentum to Absorptive Particles from a Laser Beam with a Phase Singularity”. In: *Physical Review Letters* 75.5 (1995), pp. 826–829. DOI: 10.1103/PhysRevLett.75.826.
- [15] L. Paterson et al. “Controlled Rotation of Optically Trapped Microscopic Particles”. In: *Science* 292.5518 (2001), pp. 912–914. DOI: 10.1126/science.1058591.
- [16] Alexander B. Stilgoe, Timo A. Nieminen, and Halina Rubinsztein-Dunlop. “Controlled transfer of transverse orbital angular momentum to optically trapped birefringent microparticles”. In: *Nature Photonics* 16.5 (2022), pp. 346–351. DOI: 10.1038/s41566-022-00983-3.
- [17] Graham Gibson et al. “Free-space information transfer using light beams carrying orbital angular momentum”. In: *Optics Express* 12.22 (2004), pp. 5448–5456. DOI: 10.1364/OPEX.12.005448.
- [18] Jian Wang et al. “Terabit free-space data transmission employing orbital angular momentum multiplexing”. In: *Nature Photonics* 6.7 (2012), pp. 488–496. DOI: 10.1038/nphoton.2012.138.
- [19] Nenad Bozinovic et al. “Terabit-Scale Orbital Angular Momentum Mode Division Multiplexing in Fibers”. In: *Science* 340.6140 (2013), pp. 1545–1548. DOI: 10.1126/science.1237861.
- [20] Giuseppe Vallone et al. “Free-Space Quantum Key Distribution by Rotation-Invariant Twisted Photons”. In: *Physical Review Letters* 113.6 (2014), p. 060503. DOI: 10.1103/PhysRevLett.113.060503.
- [21] Alicia Sit et al. “High-dimensional intracity quantum cryptography with structured photons”. In: *Optica* 4.9 (2017), pp. 1006–1010. DOI: 10.1364/OPTICA.4.001006.
- [22] Fabrizio Tamburini et al. “Twisting of light around rotating black holes”. In: *Nature Physics* 7.3 (2011), pp. 195–197. DOI: 10.1038/nphys1907.
- [23] Dong-Sheng Ding et al. “Single-photon-level quantum image memory based on cold atomic ensembles”. In: *Nature Communications* 4.1 (2013), p. 2527. DOI: 10.1038/ncomms3527.
- [24] A. Nicolas et al. “A quantum memory for orbital angular momentum photonic qubits”. In: *Nature Photonics* 8.3 (2014), pp. 234–238. DOI: 10.1038/nphoton.2013.355.
- [25] Lluís Torner, Juan P. Torres, and Silvia Carrasco. “Digital spiral imaging”. In: *Optics Express* 13.3 (2005), pp. 873–881. DOI: 10.1364/OPEX.13.000873.
- [26] Néstor Uribe-Patarroyo et al. “Object Identification Using Correlated Orbital Angular Momentum States”. In: *Physical Review Letters* 110.4 (2013), p. 043601. DOI: 10.1103/PhysRevLett.110.043601.
- [27] Miles Padgett and Richard Bowman. “Tweezers with a twist”. In: *Nature Photonics* 5.6 (2011), pp. 343–348. DOI: 10.1038/nphoton.2011.81.
- [28] K. Y. Bliokh et al. “Spin-orbit interactions of light”. In: *Nature Photonics* 9.12 (2015), pp. 796–808. DOI: 10.1038/nphoton.2015.201.

- [29] Timo A. Nieminen et al. “Angular momentum of a strongly focused Gaussian beam”. In: *Journal of Optics A: Pure and Applied Optics* 10.11 (2008), p. 115005. DOI: 10.1088/1464-4258/10/11/115005.
- [30] Oscar Javier Gomez Sanchez et al. “Enhanced Photo-excitation and Angular-Momentum Imprint of Gray Excitons in WSe₂ Monolayers by Spin–Orbit-Coupled Vector Vortex Beams”. In: *ACS Nano* 18.17 (2024), pp. 11425–11437. DOI: 10.1021/acsnano.4c01881.
- [31] Konstantin Y. Bliokh and Franco Nori. “Transverse and longitudinal angular momenta of light”. In: *Physics Reports* 592 (2015), pp. 1–38. DOI: 10.1016/j.physrep.2015.06.003.
- [32] Stephen M. Barnett and L. Allen. “Orbital angular momentum and nonparaxial light beams”. In: *Optics Communications* 110.5 (1994), pp. 670–678. DOI: 10.1016/0030-4018(94)90269-0.
- [33] David J. Griffiths. *Introduction to Electrodynamics*. 4th ed. Cambridge: Cambridge University Press, 2017. ISBN: 978-1108420419.
- [34] David L. Andrews and Mohamed Babiker, eds. *The Angular Momentum of Light*. Cambridge: Cambridge University Press, 2012. DOI: 10.1017/CBO9780511795213.
- [35] Safa O. Kasap and Ravindra Kumar Sinha. *Optoelectronics and Photonics: Principles and Practices*. 2nd ed. Boston: Pearson, 2013. ISBN: 9780273774174.
- [36] Frank L. Pedrotti, Leno M. Pedrotti, and Leno S. Pedrotti. *Introduction to Optics*. 3rd ed. Cambridge: Cambridge University Press, 2017.
- [37] L. C. Dávila Romero, D. L. Andrews, and M. Babiker. “A quantum electrodynamics framework for the nonlinear optics of twisted beams”. In: *Journal of Optics B: Quantum and Semiclassical Optics* 4.2 (2002), S66–S72. DOI: 10.1088/1464-4266/4/2/370.
- [38] Kayn A. Forbes, Dale Green, and Garth A. Jones. “Relevance of longitudinal fields of paraxial optical vortices”. In: *Journal of Optics* 23.7 (2021), p. 075401. DOI: 10.1088/2040-8986/abff96.
- [39] V. V. Klimov et al. “Mapping of focused Laguerre-Gauss beams: The interplay between spin and orbital angular momentum and its dependence on detector characteristics”. In: *Physical Review A* 85.5 (2012), p. 053834. DOI: 10.1103/PhysRevA.85.053834.
- [40] Shojiro Nemoto. “Nonparaxial Gaussian beams”. In: *Applied Optics* 29.13 (1990), pp. 1940–1946. DOI: 10.1364/AO.29.001940.
- [41] Kristan Bryan Simbulan et al. “Selective Photoexcitation of Finite-Momentum Excitons in Monolayer MoS₂ by Twisted Light”. In: *ACS Nano* 15.2 (2021), pp. 3481–3489. DOI: 10.1021/acsnano.0c10823.
- [42] David J. Griffiths and Darrell F. Schroeter. *Introduction to Quantum Mechanics*. 3rd ed. Cambridge: Cambridge University Press, 2018.
- [43] Lorenz Maximilian Schneider et al. “Optical dispersion of valley-hybridised coherent excitons with momentum-dependent valley polarisation in monolayer semiconductor”. In: *2D Materials* 8.1 (2021), p. 015009. DOI: 10.1088/2053-1583/abff94.
- [44] David H. McIntyre, Janet Tate, and Chad A. Staton. *Quantum Mechanics: A Paradigms Approach*. Boston: Pearson, 2012. ISBN: 978-0321765796.

- [45] Robin Bajaj et al. “Symmetries in zero and finite center-of-mass momenta excitons”. In: *Physical Review B* 112 (2025), p. 245127. DOI: 10.1103/hv2z-xx4.
- [46] Jonah B. Haber et al. “Maximally localized exciton Wannier functions for solids”. In: *Physical Review B* 108 (2023), p. 125118. DOI: 10.1103/PhysRevB.108.125118.
- [47] Diana Y. Qiu, Ting Cao, and Steven G. Louie. “Nonanalyticity, Valley Quantum Phases, and Lightlike Exciton Dispersion in Monolayer Transition Metal Dichalcogenides: Theory and First-Principles Calculations”. In: *Physical Review Letters* 115.17 (Oct. 2015), p. 176801. DOI: 10.1103/PhysRevLett.115.176801.



Appendix

A.1 Wave equations from Maxwell's equations in free space

In the electrodynamics framework, Maxwell's equations in vacuum can be written as

$$\nabla \cdot \mathbf{E}(\mathbf{r}, t) = \frac{\rho}{\epsilon_0} = 0, \quad (\text{A.1.1})$$

$$\nabla \cdot \mathbf{B}(\mathbf{r}, t) = 0, \quad (\text{A.1.2})$$

$$\nabla \times \mathbf{E}(\mathbf{r}, t) = -\frac{\partial}{\partial t} \mathbf{B}(\mathbf{r}, t), \quad (\text{A.1.3})$$

$$\nabla \times \mathbf{B}(\mathbf{r}, t) = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}(\mathbf{r}, t) = \frac{1}{c^2} \frac{\partial}{\partial t} \mathbf{E}(\mathbf{r}, t). \quad (\text{A.1.4})$$

Here $\nabla \equiv \frac{\partial}{\partial x} \hat{\mathbf{x}} + \frac{\partial}{\partial y} \hat{\mathbf{y}} + \frac{\partial}{\partial z} \hat{\mathbf{z}}$ represents the gradient operator along the three spatial coordinates. These equations can also be expressed in terms of electromagnetic potentials, which will be useful later when discussing light-matter interaction. First, the divergence-free condition of Eq. (A.1.2) allows the magnetic field to be written in terms of a **vector potential** $\mathbf{A}(\mathbf{r}, t)$ as

$$\mathbf{B}(\mathbf{r}, t) = \nabla \times \mathbf{A}(\mathbf{r}, t). \quad (\text{A.1.5})$$

Substituting this expression into Faraday's law (Eq. (A.1.3)) yields a quantity whose curl vanishes. This allows the introduction of a **scalar potential** $\phi(\mathbf{r}, t)$ such that

$$\nabla \phi(\mathbf{r}, t) = \left(\mathbf{E}(\mathbf{r}, t) + \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}, t) \right). \quad (\text{A.1.6})$$

Therefore, the electric field can be expressed in terms of the scalar and vector potentials as

$$\mathbf{E}(\mathbf{r}, t) = -\nabla \phi(\mathbf{r}, t) - \frac{\partial}{\partial t} \mathbf{A}(\mathbf{r}, t). \quad (\text{A.1.7})$$

This expression automatically satisfies both Eq. (A.1.2) and Faraday's law (Eq. (A.1.3)). Finally, Gauss's law (Eq. (A.1.1)) and Ampère's law with Maxwell's correction (Eq. (A.1.4)) can be rewritten as two coupled wave equations for the electromagnetic potentials:

$$\nabla^2 \phi(\mathbf{r}, t) + \frac{\partial}{\partial t} (\nabla \cdot \mathbf{A}(\mathbf{r}, t)) = 0 \quad (\text{A.1.8})$$

$$\nabla^2 \mathbf{A}(\mathbf{r}, t) - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{A}(\mathbf{r}, t) - \nabla \left(\nabla \cdot \mathbf{A}(\mathbf{r}, t) + \frac{1}{c^2} \frac{\partial}{\partial t} \phi(\mathbf{r}, t) \right) = 0. \quad (\text{A.1.9})$$

A.2 scalar Helmholtz equation

Begin with the Lorenz gauge, we assume a **monochromatic wave** ansatz, $\mathbf{A}^L(\mathbf{r}, t) = \text{Re} [\mathbf{A}^L(\mathbf{r})e^{-i\omega t}]$, which transforms Eq. (2.8) into the vector Helmholtz equation

$$\nabla^2 \mathbf{A}^L(\mathbf{r}) + q_0^2 \mathbf{A}^L(\mathbf{r}) = 0. \quad (\text{A.2.1})$$

Next, we introduce the **transverse wave** ansatz, $\mathbf{A}^L(\mathbf{r}) = \hat{\varepsilon} A^L(\mathbf{r})$, where $\hat{\varepsilon}$ is a polarization unit vector. Substituting this form into Eq. (A.2.1) reduces the equation to the scalar Helmholtz equation

$$\nabla^2 A^L(\mathbf{r}) + q_0^2 A^L(\mathbf{r}) = 0. \quad (\text{A.2.2})$$

